

Operationalizing Screening Off for Rainforest Admission

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Abstract

Scale-relative structural realism allows that real patterns may occupy more than one level of ontology, but the admittance criteria for those levels remain underdetermined. Franklin and Robertson (2024) offer the most developed recent proposal: emergence, understood as screening off plus novelty, earns an entity its place in the rainforest. Their criterion is the right one, but its stated test underdetermines it. Conditional independence admits an observational reading, on which micro-detail is redundant across the system's actual trajectory, and a modal reading, on which that redundancy survives admissible intervention. They coincide in the central cases and diverge where apparent macro-autonomy is borrowed from off-partition structure. A macro-variable driven by a shared upstream process can satisfy their conditions in full, novelty included, while distinct micro-realizations of its macrostate come apart under perturbation. On the modal reading, screening off and transition autonomy are not two conditions but one: to screen off micro-detail from the macro-future under admissible perturbation is to carry autonomous transition structure at that grain. Closure is the admissibility-indexed test that recovers this modal content. A candidate qualifies when, for a fixed regime, horizon, and admissible intervention class, macrostate information suffices for macro-transitions. Strong lumpability is the exact benchmark; non-ideal cases are graded by leakiness, which must itself be indexed to the intervention class rather than to the observed distribution. Interventions serve throughout as diagnostic probes rather than as metaphysical primitives. The contribution is a correction to how an existing filter is read and applied rather than a rival to it.

Keywords: real patterns, rainforest realism, closure, strong lumpability, structural realism, interventionism

1 Introduction: Closure and the Admittance Problem

Lewis observed that we have no principled reason to deny existence to the mereological sum of “the right half of my left shoe plus the Moon plus the sum of all Her Majesty’s ear-rings,” however sensible it is to ignore such things in ordinary thought (Lewis 1986, p. 213). Fair enough. But consider the difference between that composite and a hurricane. Knowing a hurricane’s pressure organization and rotational structure tells you what it will do next without tracking every molecule. The shoe-moon composite carries no comparable transition structure; predicting its future requires independently tracking each component. The question is not whether both exist. The question is which patterns carry autonomous macro-dynamics and therefore deserve robust commitment.

Scale-relative structural realism says that the world is not exhausted by a single privileged descriptive level. The special sciences succeed by tracking higher-level patterns stable enough to support prediction, explanation, and intervention, and Ladyman and Ross’s rainforest realism gave this thought its most influential metaphysical form: reality is populated by real patterns at many scales, not only by microphysical furniture (Ladyman and Ross 2007). But one question remains insufficiently settled. Which patterns earn admission into that rainforest, and which should be treated more cautiously as useful summaries, temporary proxies, or dynamically idle aggregates?

Franklin and Robertson offer the most developed recent answer (Franklin and Robertson 2024). Emergence earns rainforest admission when the macro-variable renders micro-details conditionally irrelevant (screening off) relative to the explanandum, and the macro-variable participates in a dependency profile that is not type-identical to the realizing micro-dependencies. That is a substantial advance. It captures why obvious gerrymanders and trout-turkey style composites should be excluded, while preserving the thought that higher-level ontology is earned by stable macro-level dependence rather than by intuitive naturalness alone. They explicitly cast the issue as one of the “admittance criteria for the rainforest” (Franklin and Robertson 2024, §1), and they do so in a way meant to remain compatible with theoretical reducibility rather than to revive a strong irreducibility test.

Screening off plus novelty is indispensable, but admits two readings of conditional independence, observational and modal. No published test in this literature fixes the modal reading, and the two readings return opposite verdicts on candidates the debate cares about. A candidate macro-variable deserves robust commitment only under the modal reading: it must carry its own transition structure under a fixed regime, horizon, and admissible intervention class. In exact Markov settings, strong lumpability is the benchmark realization of that condition. In non-ideal settings, closure is graded via leakiness and convergent diagnostics under fixed constraints, with explicit downgrade rules when verdicts are fragile. A candidate can satisfy observational screening-off conditions and enter into robust, law-like macrodependencies while still borrowing its apparent autonomy from upstream structure or from hidden micro-identity. Closure supplies the admissibility-indexed selector that forces the disambiguation.

The dialectical structure is cumulative rather than competitive. Dennett’s real patterns identify candidate structures through compression and projectibility (Dennett

1991); Franklin and Robertson supply the admittance filter by requiring screening off plus novelty (Franklin and Robertson 2024); the criterion developed here specifies which reading of that filter earns admission and provides the test that discriminates it. The candidates it withholds commitment from are those that pass the filter observationally while remaining parasitic on off-partition structure, whether hidden micro-identity, a shared upstream cause, or a fast hidden relaxation process. The aim is to sharpen rainforest realism from inside its own commitments rather than to displace them.

The wedge that forces the further step is a class of parasitism cases in which observational success masks off-partition dependence. The canonical example is the latent-driver proxy: a macro-variable aggregating two subsystems driven by a shared upstream process can satisfy observational screening off because the driver is a common cause of the proxy’s present and its future, so that the proxy’s current state carries exactly the upstream information its next state turns on. Under passive observation the proxy looks transition-autonomous. Perturb the proxy directly with an admissible intervention while leaving the drivers intact, and hidden heterogeneity reappears: within-class micro-differences now matter for the ensuing macro-transition. Section 3.2 makes this case explicit as a countermodel rather than an intuition, since the argument turns on the claim that a candidate can satisfy the stated admittance conditions in full and still fail transition autonomy. A second case, hidden-feedback parasitism (§3.2.2), shows the same failure signature from a different source, when fast off-partition dynamics rather than a shared upstream driver sustain the apparent autonomy. The full development in §3.2 shows why these cases decide the criterion.

Throughout this paper, *transition autonomy* names the property the criterion tests: the candidate partition’s own dynamics close under admissible perturbation, so within-class micro-detail adds no predictive power. *Gerrymandered* then means dynamically incoherent: a partition that fails transition autonomy by requiring persistent within-class micro-bookkeeping, not merely one that is intuitively disjunctive or aesthetically odd. *Parasitic* means that a macro-variable’s apparent autonomy is borrowed from upstream structure rather than realized in the candidate partition’s own transition organization.

Three claims of different strength are in play, and readers who grant one but not the next can take the paper’s contribution in corresponding part. The weakest is that closure is a necessary anti-gerrymandering constraint on serious macro-ontology claims. The middle claim is that, given the structural realist and interventionist commitments stated below, closure under admissible intervention is the appropriate criterion for robust macro-level commitment. The strongest is programmatic: rainforest realism is naturally read as a structured space of computationally closed coarse-grainings rather than as a binary inventory of what is, or is not, real. Sections 2 through 4 develop the first two claims; the structured-space reading is offered as a programmatic upshot in Section 6, with the formal richness of that space in any given domain to be cashed out as the closure discipline matures.

The argument is philosophical, not a methods paper in disguise. It provides a criterion and a disciplined protocol for applying it, while estimator selection, finite-sample behavior, and domain-specific implementation remain downstream methodological tasks.

1.1 Novelty and Positioning

The novelty claim has three parts.

1. Recovering screening off’s modal content: the contribution is not a second condition layered on top of screening off but a correction to how screening off is read. Conditional independence admits an observational reading, on which micro-detail is redundant given the macrostate across the system’s actual trajectory, and a modal reading, on which that redundancy survives admissible intervention. These readings coincide in the central cases and come apart exactly where apparent macro-autonomy is borrowed from off-partition structure. Franklin and Robertson’s criterion is retained in full; the claim is that their stated test, like the screening-off diagnostics it draws on, fixes only the observational reading, whereas the reading that carries ontological weight is the modal one. On the modal reading, screening off and transition autonomy are not two properties but one: to screen off micro-detail from the macro-future under admissible perturbation just is to carry autonomous transition structure at that grain. Closure is the admissibility-indexed test that recovers this modal content and thereby distinguishes autonomous macro-variables from parasitic ones. Their account is valuable precisely because it frames rainforest admission without treating reducibility by itself as disqualifying (Franklin and Robertson 2024). The reading is not a peculiarity of one paper. Meyer generalizes the same package beyond the special sciences and states the point at issue explicitly, defining dependence as unconditional relevance and *autonomy* as conditional irrelevance, $P(E_2 \mid E_1 \ \& \ B) \approx P(E_2 \mid E_1)$ (Meyer 2026, p. 19). Autonomy is there identified with a probabilistic condition on an unindexed distribution, which is precisely the identification this paper argues underdetermines what it is meant to secure.
2. Beyond formal closure results alone: recent work distinguishes informational, causal, and computational closure as formal diagnostics within the formal settings studied (Rosas et al. 2024), but does not say when those diagnostics warrant ontological commitment or when commitment should be withdrawn. Those diagnostics are moreover defined over the observed process, and their authors are explicit that the counterfactual reading requires the underlying system to be fully observed (Rosas et al. 2024, Sec. III), which is why they secure the observational reading rather than the modal one on their own. The criterion here uses those results for a more explicit philosophical purpose: a condition on robust macro-level standing with explicit intervention indexing, graded verdicts, and downgrade conditions.
3. Beyond effective realism without downgrade discipline: recent work in ontic structural realism allows regime-dependent, scale-relative ontology at the level of a metaphysical package rather than an individuation criterion (Ladyman and Lorenzetti 2024), but leaves open what operationally distinguishes robust admission from cases that should be downgraded or withheld. Closure supplies that discipline.

The positioning is deliberately dialogical. With Dennett, the paper keeps compression realism while refusing to let compression settle admittance by itself (Dennett 1991). With Franklin and Robertson, it agrees that screening off and novelty are indispensable but argues that they do not yet separate autonomous from parasitic

macro-variables (Franklin and Robertson 2024). With Ladyman and Ross, it keeps the rainforest ambition while insisting on an explicit transition criterion for robust macro-level commitment (Ladyman and Ross 2007). And with Ladyman and Lorenzetti, it takes ontology to be scale-relative and regime-dependent while remaining answerable to objective modal structure (Ladyman and Lorenzetti 2024); the present paper then uses interventions as diagnostics of structural autonomy rather than as metaphysical primitives. One consequence of intervention indexing is that the closure criterion can diverge from purely observational diagnostics: a macro-variable can achieve predictive sufficiency by tracking an off-partition driver rather than by instantiating autonomous transition structure (§3.2).

Several neighboring traditions are engaged where they become dialectically active: Simon and Ando on near-decomposability and aggregation (Simon 1962; Ando and Simon 1961), Wimsatt on non-aggregativity and reductionist bias (Wimsatt 2000), mechanistic accounts of organized explanation (Machamer et al. 2000; Craver 2007), Rosas et al. on formal closure diagnostics (Rosas et al. 2024), causal-emergence work (Hoel et al. 2013; Dewhurst 2021), and recent efforts to tighten or reframe pattern realism and adjacent structuralist debates (Millhouse 2022; Wallace 2024; Jiang 2024, 2025; Ladyman 2026; Meyer 2026). Each is addressed at the point where its specific contribution bears on the argument. The narrower claim is that even the strongest of these views leaves open an admittance question that closure helps make more explicit and more operational.

1.2 Intervention, Structure, and the Causation Worry

Ladyman and Ross are skeptical that causation belongs in fundamental metaphysics, and recent effective ontic structural realism continues that broadly structuralist posture (Ladyman and Ross 2007; Ladyman and Lorenzetti 2024). By contrast, the criterion defended here is explicitly intervention-indexed. That can look like a retreat from structural realism to a Woodwardian metaphysics of causation. It is not.

Intervention does not function here as an ontological primitive. It functions as a diagnostic probe for structural autonomy. Admissible intervention is the most disciplined test of whether a candidate macro-partition preserves stable relational organization under perturbation. What the criterion tracks is still structure. The intervention test asks whether that structure is autonomous at the candidate grain or whether the apparent autonomy was borrowed from elsewhere. Woodwardian intervention tests typically operate within an accepted variable scheme; the criterion here asks the prior question of when a candidate coarse-graining has earned the right to figure as a macro-variable at all.

The proposal is compatible with effective ontic structural realism because Ladyman and Lorenzetti explicitly allow ontology to be regime-dependent and scale-relative while remaining realist about the structures discovered there (Ladyman and Lorenzetti 2024). The present account agrees, but adds a sharper admittance condition: a scale-relative structure earns robust ontological standing when its transition profile is stable under the admissible perturbations that define the regime. In that sense, intervention is used here to supply operational discipline for effective realism rather than to compete with it. Their claim is not that ontology collapses into convenience,

but that ontology is always effective ontology at some scale and under some modal profile (Ladyman and Lorenzetti 2024); closure adds a stricter account of when such scale-relative commitment should be robust, qualified, or withheld.

The paper requires intervention-invariance as evidence that a candidate partition tracks stable organization across physically real, regime-preserving perturbations to boundary conditions, couplings, or control channels, not interventions as metaphysical glue. That is a methodological route to structural stability, not a retreat to fundamental causal ontology.

There is a principled reason to resist intervention-indexing that deserves a direct answer, since it is not the familiar complaint that interventions are merely difficult to perform. Meyer defends the conditional-probability approach partly on the grounds that it avoids spatiotemporal presuppositions and preserves metaphysical neutrality, which matters where the spatiotemporal and causal structure that interventionist tests presuppose is itself what is in question, as in quantum gravity (Meyer 2026). That argument is a good one, and the reply is a concession rather than a rebuttal. Where no admissible intervention class can be specified, the criterion does not deliver a robust verdict; it delivers an indeterminate one, which is the downgrade discipline working as intended rather than failing. The scope claimed here is induced closure in systems that have a control ecology (§1.3), and in that scope the observational reading is not neutral between autonomous and parasitic candidates, which is what the argument below turns on. Meyer’s desideratum and this criterion are therefore answering different questions: his asks what travels into domains where intervention talk may not apply, and this one asks what settles admission where it does. The cost of the neutral reading is that it cannot discriminate exactly where discrimination is possible.

1.3 Scope and Non-Claims

Closure comes in two forms, stipulated and induced. Stipulated closure holds when autonomy is fixed by constitutive rules, as in chess or formal arithmetic; once the rules are set, transitions and consequences follow by construction. Induced closure holds when autonomy is discovered through dynamical screening-off in the physical world. The restriction to induced closure is principled rather than ad hoc. Induced closure asks whether transition autonomy is sustained by physical and organizational structure independent of whether anyone stipulates that it should be; stipulated closure asks only whether a consequence-relation holds within a rule-system. Institutional kinds are hybrid, combining constitutive rules with an implementation substrate, and the criterion tests whether the induced component sustains the stipulated one under admissible perturbation. That is why a currency can fail closure while its legal-tender rules persist unchanged: what degrades is the induced infrastructure of enforcement and settlement, not the stipulated rule. The scope here is accordingly restricted to induced closure in spatiotemporal systems, leaving aside broader debates about abstract objects and a complete metaphysics of levels; where those debates arise, they are handled only to protect the central claim from predictable misreadings. Still, the criterion is stated over state-transition structure: spatiotemporal dynamics are the paradigm case, not the only intelligible format in which closure questions can be posed.

The criterion is also not offered as a complete account of objecthood. It answers one question: whether a candidate carries autonomous transition structure at its grain. Several neighboring accounts answer different questions about the same entities, and they are best read as complements rather than rivals. Contextual emergence shares the regime-relativity defended here, since it treats higher-level descriptions as licensed by contingent contexts that lower-level description does not itself supply (Bishop and Atmanspacher 2006). Process ontology shares the view of objects as persistent organizations of change rather than static bundles of matter (Dupré and Nicholson 2018; Jiang 2025). Structurally oriented hylomorphism is compatible to the extent that “form” is read as the organizing structure a substrate realizes (Jaworski 2016). Accounts that tie objecthood to physical binding describe one important way closure is generated in some physical domains rather than a general condition on it. Functional and historical accounts of artifacts explain why something is classified and individuated as a hammer or a chair (Thomasson 2007), which is a different question from what autonomous dynamical structure it realizes. These accounts can explain how closed structure is realized, stabilized, classified, or individuated. What closure contributes is the dynamical component that any of them presupposes when they treat a higher-level entity as doing something in its own right: if genuinely autonomous macro-structure is present, that autonomy shows up in the transition structure.

For the same reason, the criterion is not an answer to the special composition question of when some parts compose a whole (van Inwagen 1990; Lewis 1986). It asks when a coarse-graining carries its own dynamics, and the lower-level constituents figure only as what realizes that dynamics. This leaves the view compatible with several positions on composition, including structuralist views on which talk of objects is dispensable in favor of structure (French 2014). Someone who accepts all the transition facts the criterion identifies but declines to call the resulting structures “objects” disagrees with the present account mainly about the word. The claim defended here is about robust macro-level commitment, and nothing in the argument turns on the label.

Methodologically, the paper does not depend on the strongest anti-analytic rhetoric sometimes associated with naturalized metaphysics. The claim is narrower: closure is a usable philosophical constraint that can be motivated by dynamical and coarse-graining practice, whether or not one accepts every meta-metaphysical commitment in the broader *Every Thing Must Go* (ETMG) program (Ladyman and Ross 2007).

The contrast class is correspondingly focused. The paper rejects the view that compression and prediction are sufficient for realist commitment, but it also rejects a fallback to utility alone. The claim defended here is closure realism: compression and projectibility identify serious candidates, while robust macro-level commitment requires transition autonomy under fixed regime, horizon, and admissible intervention class. The paper has three argumentative layers: Section 2 states the criterion and its admissibility discipline; Section 3 gives the exact benchmark in Markov settings; Section 4 extends the same criterion to non-ideal settings as a graded protocol. Section 5 addresses the strongest objections. Section 6 summarizes the gains and limits.

2 The Criterion: Closure Under Admissible Intervention

2.1 Criterion in Plain Language

Closure can be stated without heavy formalism. A candidate macro-variable passes when knowing its current macrostate is enough to determine its macro-future over a specified horizon and intervention class. If two microstates inside one macrostate produce different macro-transitions, closure fails at that grain.

Static patterns are not exceptions; they represent dynamical regimes where micro-fluctuations are screened off rather than absent, so persistence is itself a type of macro-transition that falls squarely within the criterion. What matters is transition sufficiency, not descriptive fit. A representation can summarize trajectories elegantly and still fail to support robust macro-level commitment if it requires persistent within-class micro-bookkeeping to preserve forecast quality.

Since the formal machinery evaluates partitions of state spaces rather than physical things directly, an explicit bridge is needed. On the structural realist commitments defended below (§2.10), realist commitment at the macro level tracks stable relational and causal organization at the relevant grain; the partition identifies that organization.

The words “test” and “criterion” can suggest that closure is something investigators do. It is not. Whether a partition carries autonomous transition structure under a given regime, horizon, and intervention class is a fact about the system, and it holds or fails whether or not anyone has found it. The protocol developed below is how that fact is discovered and how claims about it are kept honest. So when the paper requires that regime, horizon, and admissible class be declared before scoring, the requirement protects the evidence. It does not make the regime. A regime is significant because the world supports autonomous transition structure there. Investigators can be wrong about where such regimes lie, but their existence is not up to anyone. This separation is made explicit in the verdict structure of §4.8, where a candidate’s closure profile is distinguished from the evidential grade of our access to it.

Closure also does not assign one privileged partition to each physical system. The same substrate can support several closed coarse-grainings in overlapping regimes, and each can earn standing in its own regime. Ordinary kind terms need not track these structures one to one. A conceptual illustration makes the point. Suppose a lump of wood and metal formed by accident and, under the interventions relevant to striking and pulling nails, behaves exactly as a manufactured hammer does. In that regime it realizes the same transition structure as the hammer. Whether it is a hammer in the artifact-theoretic sense, given that nobody made it for that purpose, is a separate question that ordinary language bundles with the first. The criterion answers only the first.

Four levels need to be kept apart to avoid recurring misunderstandings.

1. World dynamics: the implemented process itself.
2. Pattern type: stable transition structure supported by that process.
3. Pattern token: a concrete instance under specific boundary conditions.
4. Representation: a model, equation, classifier, or narrative that tracks the pattern.

A representation can fail while the pattern type remains real. A token can fail while the type remains robust. A token can also succeed while the type is weak, for example in narrow calibration windows. Closure claims here are primarily type-level claims under specified regimes, then secondarily claims about token reliability under perturbation.

Three quick clarifications forestall predictable misreadings. Closure is not a test for every sense in which a pattern might be real; it is a test for robust macro-level standing under declared constraints, so weaker senses of pattern reality remain available to candidates that do not meet it. Closure is not a demand for perfect transition autonomy; the graded verdict structure handles partial cases through qualified and indeterminate categories. And closure is not a replacement for the special-science kinds it evaluates; most will earn qualified rather than robust standing, and that is the intended output rather than a failure of the criterion.

2.2 Which Reading of Screening Off Closure Secures

A natural rejoinder is that screening off, properly understood, already excludes parasitic candidates, so closure merely renames an existing criterion. The concession is worth making directly. Closure is not a rival to screening off, and the argument does not require that Franklin and Robertson overlooked causation or meant something weaker than they said. What closure supplies is not a further condition but a determinate reading of an existing one, together with the test that secures it. That is a substantive contribution only because the two readings come apart, and come apart on candidates that matter.

1. Weak reading: observational conditional irrelevance. A variable's value renders further micro-detail probabilistically redundant for the target given the system's actual dynamics. This is a statement about the observational joint distribution, and it is what statistical screening-off diagnostics test.
2. Strong reading: stability under admissible intervention. The same conditional independence continues to hold under the interventional distribution generated by admissible perturbations that target the candidate partition directly, so that screening off is tracking autonomous structure rather than a shared source.

The claim here is that established screening-off diagnostics generally secure only the weak reading, while the strong reading is what warrants robust macro-level commitment. Closure operationalizes the strong reading by adding the admissible intervention class as a parameter of the test. The two readings agree in the central cases and diverge when upstream structure lies outside the modeled state space, or when admissible interventions target the candidate partition directly while leaving that structure intact. The latent-driver proxy developed in §3.2 is the canonical wedge; the hidden-feedback case (§3.2.2) supplies a second wedge whose borrowed stability has a different source, so the divergence is not tied to one peculiar configuration.

The reply to the rejoinder is therefore to accept it and press it further, and the crux can be stated first: a reading that no stated test discriminates cannot adjudicate admission, and the parasitism cases show that the undiscriminated readings return

opposite verdicts on candidates the debate cares about. Screening off properly understood is modal screening off, and modal screening off is transition autonomy: there are not two properties here but one, described once in the vocabulary of conditional independence and once in the vocabulary of dynamics. So the concession does not make the criterion idle. The contribution is the test and the discipline that goes with it, not the discovery of a new property. The intervention-indexed reading is offered as one principled way of making the intended modal content operationally discriminating, which is why the argument is best understood as an explication of rainforest admission from within rather than an objection to it.

2.3 Formal Statement

Let micro-process be X_t and candidate macro-process be $Z_t = g(X_t)$. For horizon L , closure asks whether Z_t is sufficient for forecasting Z_{t+1}^L under a fixed regime and admissible intervention class.

The horizon object here is the L -step path distribution, not only one-step prediction. One-step tests can be used as diagnostics, but realist claims are indexed to the declared horizon target.

Informationally, the predictive side asks whether adding X_t beyond Z_t yields material gain for macro-future prediction. Interventionally, the causal side asks whether macro-transition laws remain stable across admissible perturbations without needing within-class micro-identification.

These are not competing tests. They are two readings of the same target: transition autonomy at the macro level.

2.4 Why This is More Than Compression

Compression is necessary but not sufficient for robust macro-level commitment, because a compressed code can hide transition-relevant heterogeneity and still score well on narrow tasks. Closure blocks that loophole: does hidden heterogeneity matter for onward macro-behavior?

Closure therefore functions as an anti-gerrymandering condition, rewarding coarse-grainings that carry stable transition structure under declared constraints rather than convenient coding alone.

Put differently, the account accepts Dennett's claim that compression success is evidence of objective patterning while denying that compression success is the full ontological test (Dennett 1991), and it accepts Ladyman and Ross's concern that real patterns should be projectible and structurally disciplined while adding an explicit transition criterion for adjudicating borderline cases (Ladyman and Ross 2007). The aim is not to replace those insights but to make their realist force less permissive.

This also answers the Haugeland-style pressure point: the transition-autonomy condition determines when pattern talk has ontological force rather than merely descriptive convenience (Haugeland 1998).

2.5 Closure and the Primacy of Physics Constraint

Ladyman and Ross’s Primacy of Physics Constraint is often read negatively, as a warning against ontologies that float free of physics. But it contains a second, equally important pressure: macro-candidates must not only be physically compatible but must also avoid redundancy with lower-level description in the realism-relevant sense (Ladyman and Ross 2007). The missing question is what that non-redundancy amounts to once one gives up the idea that only the micro-level is fully real.

Closure gives that vague requirement operational content. A macro-candidate is non-redundant when micro-differences within one macroclass do not alter macro-transitions under the declared regime, horizon, and admissible intervention class. That is a stronger claim than descriptive usefulness, and a different claim from mere compression. A macro-description can be shorter, more tractable, and even explanatorily attractive while still remaining redundant in the sense that its apparent autonomy depends on continued micro-level repair.

The latent-driver pressure case brings the point home. A proxy can look perfectly projectible under passive observation because upstream structure keeps its trajectories aligned, yet turn out to be transition-parasitic: perturb the candidate directly with an admissible intervention, and within-class micro-differences matter again. The apparent macro-autonomy disappears. Closure therefore sharpens the Primacy of Physics Constraint’s non-redundancy demand by making it a question about transition structure rather than about descriptive convenience.

Read this way, closure is not an external supplement imposed on rainforest realism but a way of saying more exactly what it takes for a higher-level pattern to be physically respectable without collapsing back into lower-level bookkeeping. The realism-relevant discriminator is causal autonomy under admissible perturbation, not microphysical smallness by itself.

2.6 Admissibility and Hierarchical Evaluation

Closure is always relative to an intervention class, and the standard objection is that this invites circularity: who picks the class? The answer is procedural rather than metaphysically ambitious. Any serious macro-ontology has to answer the same admissibility question, since any view that evaluates macro-variables has to say which perturbations, regimes, and horizons are relevant. Closure does not invent that burden; it makes the corresponding choices public, comparable, and downgrade-sensitive. Domain interests help determine which control channels are worth testing, but those choices are constrained by the system’s dynamics, the target regime, and the physically available modes of perturbation. What matters for objectivity is that admissibility classes are declared in advance, cross-compared, and audited. Fragility across nearby defensible admissibility choices is itself evidence for downgrade. On this view, robust macro-level standing turns on stability of macro-transition structure across the perturbations the regime can physically sustain, not merely on fit to observed trajectories.

This procedural discipline is the framework’s main defense against gerrymandering: a candidate does not earn robust standing simply because one carefully chosen

intervention class makes it look good. The point is not to pretend that human interests disappear. It is to ensure that they cannot silently protect a favored partition from downgrade.

Admissibility is fixed upstream of commitment verdicts by three constraints:

1. Epistemic admissibility: interventions are measurable and implementable by bounded agents.
2. Dynamical admissibility: interventions preserve the target regime class.
3. Explanatory admissibility: interventions target variables with cross-context generalizability and non-trivial counterfactual reach, rather than one-off manipulations tuned to a single episode.

These constraints are physically anchored in available control channels and implementation structure, not in analyst preferences about favored ontologies. The explanatory constraint is set before closure verdicts and does not assume in advance which candidate partition will prove stable. This makes the interventionist notion of a “possible experiment” concrete and bounded rather than leaving it at conceptual possibility (Woodward 2015), and aligns with Woodward (2021)’s proportionality demand on variable choice. Explanatory admissibility is stronger when its control channels already exist in the domain and replicate across sites, weaker when success depends on narrow one-off tuning. Domain expertise shapes regime identification and control-channel selection, but that shaping is declared, auditable, and separated from partition scoring.

Pearl’s $do(x)$ surgery is defined mathematically regardless of physical feasibility, which is productive for causal inference (Pearl 2009). Robust macro-level commitment requires tighter constraints: a partition that closes only under interventions no bounded agent could implement without destroying the target regime does not support the intervention-guiding structure that warrants realist commitment. The restriction concerns which of Pearl’s outputs carry ontological weight, not the framework itself.

This connects back to the foundational claim above. The admissible class is not arbitrary: a system’s boundaries are probed constantly by naturally occurring perturbations (thermal noise, diffusion gradients, mechanical impacts), and a cell membrane screens molecular fluctuations whether or not an experimenter is watching. These perturbations are exogenous to the candidate partition even when internal to the larger system, so the test inherits the exogeneity that standard interventionist accounts demand without requiring human agency to secure it.

Evaluation order is fixed by the canonical selection discipline (§4.3): intervention classes are fixed from independent physical constraints and established control practice before closure is scored, and post hoc revisions restart the evaluation.

Taken together, the discipline can be stated as a short acceptability test. An admissibility class is acceptable only when:

1. interventions use physically available control channels;
2. interventions preserve the regime being evaluated;
3. no intervention is one-off or directly partition-protective;
4. verdicts are checked across nearby defensible alternatives.

Classes that fail any of these conditions cannot carry the evidential weight of a robust verdict.

2.7 Regime Individuation and Circularity

If regime individuation already commits to which macro-variables matter, the protocol risks circularity: closure testing presupposes regime identification, but regime identification might presuppose partition outcomes. The reply has a structural and a procedural part. Structurally, the criterion is designed to sort cases into three classes rather than to force robust verdicts everywhere. In the first class, regime specification is uncontroversial across candidate partitions, and the criterion adjudicates robustly. In the second, regime specification is contested but iteration stabilizes under the selection discipline (§4.3), and reflective equilibrium converges on a verdict. In the third, regime specification is contested and iteration does not stabilize, so the criterion issues an indeterminate verdict; this is the discipline working as intended, not the criterion failing. Procedurally, defensible regime specifications draw on quantities characterizable without presupposing which candidate partition will succeed. Temporal parameters, energy scales, flow rates, and enforcement intensities are typically identifiable through physical indicators that competing candidate partitions can agree on before either is evaluated. A running example will help here and at two later points, and it can be stated in a line. Consider two rival partitions of weekday commuter traffic in a metropolitan corridor: P_{lane} , which tracks density and flow by lane segment, and P_{tod} , which tracks flow volume in clock-time buckets calibrated to typical commuter patterns. Both are choices a real modeler might defend, and their in-sample one-step fit can be similar. The regime is characterized by clock schedules, total vehicle counts, and road network topology, all quantities that investigators favoring either partition can agree on before either closure test is run. Regime specifications of this kind are anchored in lower-level or cross-cutting physical indicators rather than in the outcome of the very test they enable. Regime specifications should be grain-neutral across competing candidate partitions: a regime can itself be a macro-level description while still not presupposing which finer-grained partition will close within it.

Independence is imperfect in some domains, particularly biology and social science, where macro-variable choice and regime characterization can be entangled (psychiatric diagnostic categories, species concepts: the variables that pick out the regime are themselves candidates whose macro-autonomy is under test). This is where the criterion earns its keep rather than stumbles: when alternative, independently motivated regime specifications produce divergent closure verdicts, the right result *is* downgraded to at most qualified, because the evidential situation does not yet support a stronger commitment. The criterion is doing its intended work when it refuses robust standing to candidates whose apparent autonomy is parasitic on the choice of regime characterization.

Take a contested case as a stress test, sketched schematically rather than settled here. Suppose major depressive disorder is approached under three independently motivated regime characterizations: a symptom-cluster regime tied to diagnostic instruments, where intervention evidence on pharmacological and psychotherapeutic

levers might support partial closure within a bounded clinical horizon; a neurobiological regime indexed to specific circuit dysfunctions, where the same label may fragment across heterogeneous mechanisms; and a social-functional regime indexed to role-impairment trajectories, tracking a third transition profile. If the case has that shape, the criterion returns qualified standing, with explicit downgrade for any commitment that depends on which characterization is accepted upstream. Whether it has that shape is a question for philosophy of psychiatry, and nothing here settles it. The point is what the criterion does with a kind whose regime is contested, not a verdict on this kind. Refusing to launder a contested kind into robust standing is the discipline working as intended, not evasion.

Regime individuation and partition evaluation are iterative rather than viciously circular: initial regime characterizations are provisional, and closure testing can refine them. The process stabilizes into reflective equilibrium in Goodman’s sense ([Goodman 1983](#)): if iteration does not stabilize across nearby defensible starting specifications, the result should be treated as indeterminate rather than forced into a robust verdict. Here stabilization means that comparative ordering across independently specified candidate partitions stops shifting under small, defensible protocol variations; it does not presuppose one privileged grain in advance. Iteration is governed by the same selection discipline (§4.3) as initial scoring: each revision restarts the evaluation rather than silently inheriting prior results. The framework sometimes relocates disagreement rather than resolving it, and that is the intended outcome in hard cases. The aim is not to guarantee a unique answer in every domain but to distinguish cases where robust commitment is warranted from cases where the evidential situation supports only qualified or indeterminate standing.

A more global version of the worry asks what selects the right contexts at all, given that some sufficiently narrow regime can be found in which almost any candidate looks stable. The answer depends on separating two kinds of robustness that the protocol checks, because they play different roles. Robustness across the admissible intervention class is part of what closure is. The criterion quantifies over every within-class realization an admissible intervention can reach (§4.2), so a candidate that holds up under some admissible perturbations and not others is not closed under that class. Robustness across nearby regimes, horizons, and modelling choices plays a different role. It is evidence that the regime under test is a real feature of the system’s dynamics rather than one tuned to a favored partition. The narrow-regime worry is answered by that second kind of robustness together with the admissibility constraints: a regime drawn so tightly that only one partition survives in it will usually fail projectibility under small context shifts (§4.4), and a regime whose boundaries were chosen by looking at closure scores violates the selection discipline. No independently privileged scientific task is needed to fix the right context. The right regimes are those in which the world supports autonomous transition structure, and the protocol is built to find them without being able to invent them.

“Regime dissolution,” the event that marks the outer edge of a regime, is not partition-relative. It is characterizable by cross-cutting physical indicators that do not presuppose the winning partition: energy-scale crossings, phase boundaries, control-response failure at established channels, and breakdowns in the separation of

timescales that defined the regime. These indicators register whether the background dynamics have changed, not whether a favored partition continues to score well.

2.8 Admissibility Disputes

Admissibility disputes are expected, especially across domains. The adjudication rule is to compare candidate intervention classes by what they physically permit and whether they preserve the same target regime. Boundary-pressure perturbations can be admissible for storm dynamics, while molecule-by-molecule remote rewriting is not. In institutional settings, changing enforcement intensity can be admissible, while instant arbitrary rewriting of all agent commitments is not. The ontological consequence is direct. A storm satisfies closure under boundary perturbations because its pressure organization and rotational structure determine macro-transitions without tracking individual molecules. Under molecule-by-molecule rewriting, the same candidate fails, because admissibility now permits interventions that reach inside the macroclass and exploit within-class differences. The candidate has not changed; the admissible control channel has, and the verdict changes with it. That is why admissibility is declared before scoring (§2.6).

The issue is distinct from regime individuation. Regime individuation asks which background dynamics define the target setting; admissibility individuation asks which intervention classes count as ontologically relevant probes within that setting. Tying explanatory admissibility to available control channels and implementation structure, rather than to interventions that happen to vindicate a favored partition, answers the circularity worry here as well.

Front-end specification of regime, target, and admissible control channels is often interest-guided in a perfectly ordinary way. That does not make closure verdicts interest-relative once those specifications are fixed and openly defended. A short template grid (boundary nudges, coarse actuator controls, policy levers) can make admissibility comparison public and repeatable in domains that lack an established control lexicon.

One guardrail is non-negotiable. Interventions that directly target preservation of partition labels, rather than system variables, are inadmissible because they trivialize closure by design. The test concerns whether transition structure is autonomous under physically meaningful controls, not whether labels can be protected by stipulation.

When two intervention classes are both admissible and target the same regime, the criterion permits plural testing. Verdicts should report which class was used and how sensitive results are across nearby defensible classes; sensitivity to a narrowly tuned class is grounds for downgrade. In the commuter corridor of §2.7, ramp metering and speed-limit controls count as admissible because they are physically available control channels in the regime, while vehicle-by-vehicle remote rewriting does not, not because that would embarrass P_{lane} but because it is not part of the regime’s actual control ecology.

2.9 Predictive and Interventional Closure

Predictive and interventional closure should be distinguished but not separated. Predictive closure concerns whether macrostate information screens off transition-relevant

micro-detail for macro-future forecasting; interventional closure concerns whether macro-transition structure remains stable when the system is probed.

The evidential relation between them is asymmetric. Strong interventional closure typically implies predictive adequacy for the same target and horizon, while predictive adequacy alone can persist in cases where interventional stability fails under distribution shift. For that reason, the criterion treats predictive evidence as important but not final.

This asymmetry also clarifies burden of proof. Claims to robust macro-level standing need either direct interventional support or compelling indirect evidence that tracks intervention-relevant invariance. Otherwise, the responsible verdict is qualified or indeterminate.

Indirect evidence can still be substantial: natural experiments, intervention-like policy shifts, quasi-experimental contrasts, and systematic failure under exogenous disruption can all reveal whether a candidate partition preserves its own transition structure across admissible changes. These sources do not collapse the distinction between predictive and interventional closure. They are useful precisely because they bear on intervention-relevant invariance rather than on in-sample fit alone.

The point matters especially in biology, economics, and the social sciences. In those domains, direct manipulation is often sparse, ethically constrained, or expensive. The ontological target nevertheless remains interventional closure. Where the evidence for that target is thin, robust verdicts should be rare. That is a feature of the graded verdict structure rather than a retreat to observational permissiveness.

2.10 Why Closure Carries Ontological Weight

A reviewer can still ask why transition autonomy should count as robust macro-level standing rather than as a mere success condition for modeling. The answer depends on the paper’s explicit commitments. Under structural realism, what ontology should track is stable relational and causal organization rather than intrinsic micro-identity as such. Under interventionism, what counts as causally relevant structure is structure that supports stable manipulation and control.

This fits the regime-dependent, scale-relative orientation defended by effective ontic structural realism. The proposal does not ask for a return to one privileged level or to thick intrinsic objects, but for a sharper criterion of when a scale-relative structure has earned commitment, and when it should be downgraded, within a broadly structural realist framework (Ladyman and Lorenzetti 2024).

The meta-philosophical stance here follows what Woodward calls the methodological path to ontology: causal-interventionist tools are used to determine what earns realist commitment, rather than seeking independent metaphysical “truth-makers” or “grounds” that stand apart from the practices of manipulation and prediction (Woodward 2015). On that path, scientific ontology disciplined by closure is the only ontology required for commitment verdicts at this grain. This is a consequence of the stated commitments rather than an independent argument against those who reject them. A grounding theorist who holds that stable dynamical autonomy under admissible intervention still stands in need of a further metaphysical foundation is not making a mistake by the lights of their own view; they are declining the methodological path

at the outset, and the disagreement is therefore about that path rather than about closure. Readers who take it will find that closure settles what commitment requires; readers who do not can still take the anti-gerrymandering result of §3.2, which does not depend on the ontological reading.

A persistent worry is that prediction and interventional control are merely epistemic lenses, leaving the underlying ontology untouched. But this confuses the test with what is being tested. Prediction and intervention are diagnostic tools that reveal pre-existing statistical structure. The conditional independencies that allow a macro-state to screen off micro-details exist whether or not anyone computes them; they are the terrain, not the lens. Locality, finite signal propagation, and thermodynamic limits create these independencies as physical facts. Our predictive models succeed when they align with this objective causal topology, and fail when they do not. Closure is therefore an ontological criterion because it identifies where the world’s phase space objectively decouples, not just where our minds prefer to chunk it. Under structural realism, such decouplings are ontological because they are the stable relational structure realist commitment was supposed to track in the first place, not because they license any particular representation. A criterion that identifies them is therefore picking out what the ontology consists in at that grain, not describing a consequence of it.

Given those commitments, closure functions not as a convenient heuristic layered on top of ontology but as the criterion that identifies when a candidate macro-description tracks stable structure at the level where explanation and intervention are being assessed. That is what lets the view remain compatible with microphysical completeness while still defending non-redundant macro commitment. The paper is not smuggling ontology out of modeling success; it is defending a specific thesis about which kind of modeling success is ontology-relevant, namely transition autonomy under admissible perturbation.

What the criterion measures is therefore not a floating label imposed from outside the formalism. A candidate’s closure profile, how much transition autonomy it has under the declared constraints, is a fact about the candidate and not about anyone’s confidence in it. The verdicts built on that profile are a further matter, since they answer to the state of the evidence as well (§4.8); the ontological claim here attaches to the profile.

2.11 Closure and Underdetermination

One remaining concern is underdetermination. Even after the criterion is stated, a reviewer may argue that many distinct coarse-grainings can be tuned to look acceptable on available data. If so, closure might seem to collapse back into model choice rather than commitment discipline.

Three questions need to be kept apart.

1. Which candidate partitions are descriptively serviceable in a dataset?
2. Which candidates remain transition-autonomous under admissible perturbation?
3. Which of those candidates remain stable under modest horizon and admissibility variation?

Underdetermination is strongest at the first question and weaker at the second. It is often weakest at the third. Many candidates can fit observed trajectories. Far fewer preserve counterfactual structure when the regime is probed. Fewer still remain stable across nearby defensible setups.

The account does not claim that every domain yields one uniquely privileged partition. It claims that realist commitments should be constrained by transition autonomy and robustness, rather than by fit alone. When underdetermination persists after those filters, the responsible result is qualified or plural commitment under transparent constraints, not forced monism and not unconstrained relativism.

When exact closure is unavailable, leakiness-centered diagnostics estimate comparative distance from closure, and commitment strength tracks the stability of that evidence.

3 Exact Benchmark: Strong Lumpability

The paper’s relation to Rosas et al. can be stated exactly. What is taken from their treatment is the formal diagnostics of informational, causal, and computational closure, and the strong-lumpability benchmark (Rosas et al. 2024). What is added here is an intervention-indexed criterion for when such diagnostics warrant robust macro-level commitment, together with downgrade rules for non-ideal cases. The formal results are read as the ideal limit of the criterion rather than as the criterion itself.

Strong lumpability is introduced as a benchmark, not as destiny: a clean exact case in Markov settings where closure can be stated and checked without ambiguity.

Let partition cells under g be macroclasses. Strong lumpability holds when microstates within the same macroclass induce identical transition probabilities to all macroclasses. This is the standard strong-lumpability condition; cf. the presentation attributed here to Kemeny and Snell (Kemeny and Snell 1960). When this condition holds, macro-transitions are autonomous by construction.

More precisely: for any two microstates x, x' in macroclass C_i and any macroclass C_j , strong lumpability requires $\sum_{y \in C_j} P(y \mid x) = \sum_{y \in C_j} P(y \mid x')$. The induced macro-kernel is then well-defined independently of representative choice, so the macro-process is itself Markov. A partition that fails this condition mixes transition-heterogeneous microstates and therefore lacks macro autonomy.

The Markov condition does not assume history is irrelevant; it assumes only that the relevant state description has been fixed. In memory-bearing systems, that description may need to be history-enriched before the closure question is asked. That does not trivialize non-Markovian cases or the formal difficulty of establishing lumpability in them; it states the standard point that the closure test applies to the least state description that properly bounds the relevant memory.

Computational mechanics puts the same point in sharper language. For any macro-process Z , two prediction machines can be defined. The ε -machine is the minimal model that predicts Z ’s future from Z ’s own past, using only macro-level information (Shalizi and Crutchfield 2001). The v -machine is the minimal model that predicts Z ’s future from the full micro-past X , using everything available (Rosas et al. 2024). Closure holds when these two machines are equivalent: the ε -machine already captures

everything the v -machine knows about macro-futures, and extra micro-information is redundant for the macro target. This gives an exact ideal where closure is not approximate. The same formal point is developed in Rosas et al.’s treatment, where strong lumpability corresponds to the exact case in which the macro-level causal-state machine coincides with the micro-conditioned prediction machine, so that no residual micro-information adds predictive leverage (Rosas et al. 2024, Thm. 3 and Prop. 4, pp. 27–28). The present account reads their formal result as the ideal limit of the criterion rather than as the criterion itself.

The limitation is equally clear: exact strong lumpability is uncommon in open, noisy, and path-dependent systems. That rarity does not undermine the criterion itself.

3.1 Minimal Formal Illustration

The anti-gerrymandering role can be shown with a minimal symbolic example. Let microstates be $\{x_1, x_2, x_3, x_4\}$ with transition rows:

$$\begin{aligned} P(x_1, \cdot) &= (\alpha, \beta, 0, 0), \\ P(x_2, \cdot) &= (\alpha, \beta, 0, 0), \\ P(x_3, \cdot) &= (0, 0, \gamma, \delta), \\ P(x_4, \cdot) &= (0, 0, \gamma, \delta), \end{aligned}$$

with $\alpha + \beta = 1$ and $\gamma + \delta = 1$.

Partition A groups $\{x_1, x_2\}$ and $\{x_3, x_4\}$. Partition B groups $\{x_1, x_3\}$ and $\{x_2, x_4\}$. In A , members of each macroclass have matching onward profiles to macroclasses, so macro-transitions are autonomous. In B , members of one macroclass differ in onward profiles whenever $\alpha \neq \gamma$, so the macro-label hides a transition-relevant difference.

Partition B can still be made predictive by adding repeated within-class bookkeeping. A partition is *high-maintenance* when reliable macro-prediction repeatedly forces refinement of within-class distinctions, so that yesterday’s macrostate labels must be supplemented by new micro-bookkeeping to retain accuracy. That is exactly the case the criterion excludes from robust macro-level standing. On a fixed sample trajectory, this bookkeeping can make Partition B look nearly as predictive as Partition A . The difference appears under admissible perturbation: A preserves macro-transition structure without renewed microstate tracking, whereas B does not.

Both partitions are definable. Only one preserves transition autonomy. Closure therefore discriminates between legitimate macro-candidates and merely codable aggregates.

3.2 How the Formal Machinery is Used Here

The mathematical property and the philosophical criterion are different things, and the difference is philosophical work the formal framework does not attempt and does not need to attempt for its own purposes. The central such addition is intervention indexing. Causal-state constructions classify histories by equivalence of future distributions under the observed process, which is primarily an observational-predictive equivalence relation (Shalizi and Crutchfield 2001). The criterion here adds an admissible intervention class as a parameter of the closure test. Realist claims are indexed to

what the system does under perturbation, not only under passive observation, which separates the criterion from a purely statistical diagnostic. Rosas et al.’s framework distinguishes informational, causal, and computational closure as complementary formal diagnostics (Rosas et al. 2024), and their Theorem 1 establishes that for spatial coarse-grainings, under the assumptions specified there, informational and causal closure coincide (Rosas et al. 2024, Thm. 1). In that framework *causal closure* is defined as equivalence between the coarse-grainings induced by a process’s ϵ -machine and its v -machine (Rosas et al. 2024, Def. 3), and *information closure* as the vanishing of $I(\vec{X}_t; \vec{Z}_{t+1}^L \mid \vec{Z}_t)$ for all L (Rosas et al. 2024, Def. 4). Both are defined over the observed process. Rosas et al. are themselves explicit about what this does and does not license: causal closure as defined there provides counterfactual guarantees only if the underlying system is fully observed, and otherwise “ought to be understood in the Granger sense, i.e. as a best guess given the available knowledge” (Rosas et al. 2024, Sec. III). The equivalence the theorem establishes is therefore one between two readings of observational structure, not between observational structure and interventional invariance under perturbation external to the modeled process. The criterion here is downstream of that result, and it takes their caveat as its point of departure. It treats those observational diagnostics as strong evidence for closure, but ties realist commitment to a further admissible-intervention stability condition and to the absence of hidden micro-identity bookkeeping. The two diagnostics agree in the central cases and can diverge when relevant upstream structure lies outside the modeled state space, or when admissible interventions target the candidate partition directly while leaving such structure intact. That divergence is not a counterexample to their result. It is the region their own full-observation proviso exempts: where the structure sustaining apparent macro-autonomy lies off-partition, the Granger reading is the operative one, and the counterfactual guarantee lapses at exactly the point where parasitism becomes possible. The latent-driver case developed below is the first wedge; the hidden-feedback case (§3.2.2) shows the same failure signature from a structurally different source.

The graded assessments the criterion issues should be read in the same spirit, with one distinction kept in view. A candidate’s closure profile reports where it sits in the space of more and less stable closure relations, and that is a claim about the candidate rather than about the state of the evidence. The commitment verdict built on the profile also answers to the evidence, so a downgrade can reflect either a fragile candidate or thin access to it (§4.8).

3.2.1 A Latent-Driver Proxy

The first case is the latent-driver proxy, where observational screening off cannot discriminate autonomous macro-structure from borrowed macro-structure. Consider a composite variable over subsystems that are pinned by an upstream driver process the partition does not track. What matters is not that a single cause synchronizes the subsystems, which is only one way the structure can be realized, but that the driver process is upstream of both the composite’s present state and its future state. The apparent dependence of the macro-variable on its own past is then underwritten by off-partition structure. Under passive observation, the composite’s macro-state

predicts its own future as well as the full micro-state does, because the driver pins the subsystems and thereby makes the proxy’s present state carry exactly the upstream information its future turns on. An ε -machine built on the composite finds no residual micro-information worth adding, and screening-off diagnostics register a clean pass. Can the screening-off-plus-novelty package handle this case?

Franklin and Robertson’s criterion asks whether the macro-variable enters into a genuine macrodependency that renders some lower-level detail conditionally irrelevant while contributing something novel at the macro level (Franklin and Robertson 2024, §3). Their screening-off condition has two parts. The dependence part, *unconditional relevance*, requires $P(A \mid LLD) > P(A)$: the lower-level description is relevant to the macro-fact. The independence part, *conditional irrelevance*, requires in its strict form that $P(A \mid B \& LLD) = P(A \mid B)$, so that all the influence of the micro-detail is mediated through the macrostate (Franklin and Robertson 2024, §3.1). Their novelty condition is more demanding than it may first appear, and a weaker reading would make the case below too easy. Novelty requires *distinct* macrodependencies: a binary star system screens off the individual stellar trajectories, yet they deny it admission, because the dependencies describing it “are just the same nomic dependencies that describe the trajectories of the two individual stars” (Franklin and Robertson 2024, §3.2). Novelty is therefore not satisfied merely by a dependence being stable at the macro grain. It requires a macrodependency of a different form from the micro-dependencies that realize it, and its mark is cross-classification, where divisions at the macro level cut across divisions below.

That is a real constraint on the pressure case, and it excludes the simplest candidates. A macro-variable that is a relabeling of a single driver inherits that driver’s law and so fails novelty for the same reason the binary star does. What follows is therefore not the simplest proxy but the simplest one that survives their whole package.

Consider an upstream driver process with two components, U^1 and U^2 , independent of each other, each a symmetric two-state Markov chain on $\{0, 1\}$ that flips with probability q , where $0 < q < 1/2$. Two subsystems A and B track their respective components. The system’s specified dynamical ecology includes one control channel, a reset line on B with indicator R_t and value C_t . The update is simultaneous, and every variable at $t + 1$ is fixed by variables at $t + 1$ or t :

$$U_{t+1}^i = U_t^i \oplus N_{t+1}^i, \quad A_{t+1} = U_{t+1}^1, \quad B_{t+1} = \begin{cases} C_{t+1} & \text{if } R_{t+1} = 1, \\ U_{t+1}^2 & \text{if } R_{t+1} = 0, \end{cases}$$

where N^1, N^2 are independent Bernoulli(q) noise terms. The micro-state is $X_t = (U_t^1, U_t^2, A_t, B_t)$, and the reset line is an exogenous control input rather than a component of X_t : the controlled process is Markov on X_t once (R_t, C_t) is fixed. The candidate macro-variable is the agreement of the two subsystems, $Z_t = A_t \oplus B_t$, taking the value 0 when they agree and 1 when they disagree.

The reset channel is part of the system’s specification rather than an intervention imagined from outside it. In the observed process $R_t = 0$ for all t , so the subsystems are pinned to their components, $A_t = U_t^1$ and $B_t = U_t^2$, and $Z_t = U_t^1 \oplus U_t^2$. The

restriction of the observed process to the synchronized manifold is therefore a fact about which channels happen to be exercised, not about what the dynamics permit.

Write $V_t = U_t^1 \oplus U_t^2$ for the driver parity and $r = q^2 + (1 - q)^2$ for the probability that the two noise terms agree. Substituting the update rule gives the identity the whole case turns on: with $R_{t+1} = 0$,

$$Z_{t+1} = V_t \oplus (N_{t+1}^1 \oplus N_{t+1}^2), \quad \text{so} \quad P(Z_{t+1} = V_t) = r.$$

The macro-variable's future is fixed by the *driver* parity, not by its own current value. The two coincide exactly when the channel is dormant, since $Z_t = V_t$ whenever $R_t = 0$. Everything below follows from that gap.

1. Unconditional relevance, taken with their indexing. Their condition ranges over the lower-level descriptions that subvene a given macrostate B and influence a macro-fact A , which together form an equivalence class (Franklin and Robertson 2024, §3.1). Fix A as $Z_{t+1} = 0$ and B as $Z_t = 0$. In the observed process the descriptions subvening B are exactly those with $V_t = 0$, namely $(0, 0)$ and $(1, 1)$, and each raises the probability of A from its unconditional value of $1/2$ to r , since $r - 1/2 = 2(q - 1/2)^2 > 0$ for $q \neq 1/2$. So $P(A \mid LLD) > P(A)$ holds across that class, which is what their condition asks. Descriptions with $V_t = 1$ lower the probability of A to $1 - r$, but they subvene $Z_t = 1$ and so belong to the class for a different macrostate.
2. Conditional irrelevance, in the strict rather than the approximate form. When $Z_t = 0$, the only micro-states the observed process ever visits are those with $V_t = 0$, and both yield $P(Z_{t+1} = 0) = r$. When $Z_t = 1$, the only visited realizations have $V_t = 1$, and both yield $1 - r$. Hence $P(Z_{t+1} \mid Z_t, X_t) = P(Z_{t+1} \mid Z_t)$ exactly. The unvisited realizations do not disturb this verdict, and it matters why. The realizations of $Z_t = 0$ with $V_t = 1$ receive probability zero in the observed process, so the conditional probability is undefined on them and the strict condition holds over them only vacuously. The condition is not violated by the heterogeneous realizations; it is silent about them, and silent about exactly the realizations on which the closure question turns. The proxy does not merely approximate screening off, in the way a deviant microstate would degrade it. It satisfies the strict condition on every realization for which the condition says anything at all.
3. Novelty, structurally rather than numerically. Z 's macrodependency relates parity classes to later parity classes, and neither its relata nor its transition partition coincide with those of either component process. The macrostate $Z = 0$ lumps $(0, 0)$ with $(1, 1)$, which cuts across the partition induced by either component's state, so the equivalence classes over which Z 's transitions are defined are not the classes over which U^1 's or U^2 's transitions are defined. This is cross-classification in Franklin and Robertson's sense, and it is why the case is not the binary star, whose centre of mass obeys the very dependencies its components obey. The numerical difference is a symptom of this rather than the argument for it: Z is a symmetric chain with persistence r against each component's $1 - q$, and $r - (1 - q) = q(2q - 1) \neq 0$ for $q \neq 1/2$. Were the parameters to coincide, the macrodependency would still not be either component's, because it would still be defined over different relata.

A defender may resist at the third condition, holding that parity is a mere algebraic consequence of two independent processes and so lacks the relevant kind of novelty. The reply is not that this reading is indefensible. It is that the reading is not the published one. If novelty requires more than cross-classification and non-identity of dependency profiles, whether organizational integration, explanatory unity, or something else, that further requirement has to be stated independently, because the probabilistic criterion as given does not supply it. A defender who adds it is conceding the paper’s point in the currency the paper asks for: the stated test underdetermines the verdict, and what settles it is a commitment the test does not contain.

So the stated implementation of the criterion returns a pass. That is weaker than saying the proxy belongs in the rainforest, and the weaker claim is the one the argument needs. What has been shown is that the conditions as stated, evaluated on the observed distribution, do not exclude this candidate. Whether Franklin and Robertson’s broader modal intentions would exclude it is exactly the question at issue, and the thesis here is underdetermination rather than refutation.

Now consider the micro-realizations of $Z_t = 0$ that the observed process never visits: those with $V_t = 1$. Take drivers at $U_t^1 = 0$ and $U_t^2 = 1$, and actuate the reset line at t with $R_t = 1$ and $C_t = 0$. Then $B_t = 0$, $A_t = U_t^1 = 0$, and $Z_t = 0$, while the drivers are untouched and $V_t = 1$. From there $P(Z_{t+1} = 0) = 1 - r$, against r for every observed realization of $Z_t = 0$, and $r \neq 1 - r$ for $q \neq 1/2$. Two micro-realizations of one macrostate carry different macro-futures, and the update rule shows why: the future tracks V_t , while the macrostate tracks V_t only for as long as the channel stays dormant.

Building the channel into the specification forecloses the natural objection, which is that the perturbation forces the system off the manifold on which the macro-variable was defined and so abandons the regime rather than testing it. It does no such thing. Actuating the reset line is an ordinary operation of the system’s own control ecology, the dynamics are unchanged throughout, the channel falls dormant again immediately, and the subsystems are repinned to their components at the next step, so the deviation is a one-step transient. The synchronized manifold was never a constraint the dynamics imposed. It was a restriction on the observed distribution produced by leaving an available channel unused, which is precisely the difference the criterion is designed to detect.

A harder version of the objection is worth meeting head on, since the countermodel invites it: any macro-variable would fail closure if one were free to posit enough internal reset channels. That is true, and it is why the admissibility constraints of §2.6 do the work they do. A channel cannot be introduced after the verdict is seen, must belong to the regime’s independently specified control ecology, must preserve the target regime, and must survive comparison with nearby defensible intervention classes. A reset line invented to embarrass a candidate fails the first and second of these immediately.

A related worry runs in the opposite direction. Organisms, artifacts, and institutions all depend on their surroundings: on gravity, on a temperature range, on nutrient flows, on supporting practices. If dependence on structure outside the partition is parasitism, the wedge seems to condemn nearly every special-science kind. It does not, because parasitism in the present sense is narrower than dependence. Background

conditions that the regime holds fixed are part of the regime’s specification. They are not borrowed by the candidate; they are the setting in which the candidate’s closure is assessed, and a change in them is a change of regime rather than a perturbation within it. The drivers in the proxy are not background conditions of this kind: they vary within the regime, and the macro-future turns on their current state. The proxy is parasitic for that reason. An admissible intervention on the candidate itself, the reset line, decouples it from the structure that sustains its apparent autonomy while leaving the regime intact. The question for any candidate is therefore not whether it depends on its environment, since everything does, but whether some admissible perturbation acting on the candidate reaches within-class realizations that the environment’s support had been hiding. An organism kept alive by a stable environment is not thereby parasitic. It would be parasitic only if an admissible perturbation acting on it showed that its apparent macro-dynamics had tracked some external driver all along.

The objection is also a useful reminder of what the countermodel does and does not show. It establishes a possibility, that the stated observational criterion can certify a candidate whose autonomy is borrowed, and it locates the condition under which this happens: some admissible channel decouples the candidate from the structure sustaining it. It shows nothing about how often that condition is met. Nothing here suggests that parasitism is widespread in the special sciences, and the argument does not need it to be. The conclusion is that the observational criterion is insufficient in principle wherever such channels exist, and whether they exist in a given domain is a question for that domain.

The proxy’s clean macro-law holds only because, in the observed process, $Z_t = 0$ guarantees that the drivers agree. The macro-variable never tracks that fact; it borrows it. Screening off quantifies over the realizations the process happens to visit, and on those realizations there is no heterogeneity to detect. Transition closure quantifies over the realizations an admissible intervention can reach, and there the heterogeneity is immediate. This is the same distinction the exact benchmark already draws: the partition is weakly lumpable, since the observed macro-process is genuinely Markov, but not strongly lumpable, since strong lumpability demands equal onward macro-transitions for every microstate in a class rather than only for those the observed distribution happens to weight.

There is also a positive way to locate the autonomy the case contains. The driver parity $V_t = U_t^1 \oplus U_t^2$ is itself a strongly lumpable coarse-graining of the driver pair: since $V_{t+1} = V_t \oplus (N_{t+1}^1 \oplus N_{t+1}^2)$ whatever the reset line does to the subsystems, every realization of (U_t^1, U_t^2) with a given parity has the same onward parity distribution, with persistence r . The autonomy in the case is therefore real; it is just located at the driver partition rather than the candidate. Z borrows V ’s closure for as long as the dormant channel keeps the two aligned. Parasitism is then not a metaphor. It names a candidate whose apparent transition structure is the closure of a different partition, one the candidate neither contains nor preserves under admissible perturbation.

The case is therefore not a verbal trick or a diagnostic failure. It is a countermodel: a candidate that satisfies unconditional relevance, strict conditional irrelevance, and

novelty, and that nonetheless fails transition autonomy. Its role is to establish possibility, not frequency. Nothing here claims that such candidates are common in the special sciences, and the model is an illustration of a structural gap rather than evidence about any actual system. What it shows is that the gap is real: the stated screening-off test cannot by itself exclude borrowed autonomy, because the borrowing is invisible in the distribution the test quantifies over.

The natural rejoinder, that screening off properly understood already excludes the proxy, was met in §2.2: the proxy is excluded on the modal reading, and closure is the test that secures that reading.

3.2.2 A Second Pressure Case: Hidden-Feedback Parasitism

A second parasitism case arises when a macro-variable appears transition-autonomous because a fast hidden process continually relaxes it toward its apparent attractor. The source of the borrowed stability is now internal and fast rather than upstream and exogenous, but the skeleton of the previous case transfers directly, so the gap is not tied to one configuration.

Retain the two subsystems A and B and the candidate $Z_t = A_t \oplus B_t$, and replace the external drivers with two fast hidden variables H^1 and H^2 that relax toward the subsystems they track. The micro-state is $X_t = (A_t, B_t, H_t^1, H_t^2)$, and the dynamics are that each subsystem's next value is its hidden variable's current value, flipped with probability q , after which the hidden variables re-equilibrate to the subsystems. Because the relaxation is fast relative to observation, the observed process never leaves the equilibrated manifold on which $H_t^1 = A_t$ and $H_t^2 = B_t$.

The verdicts are those of the previous case, and for the same reasons. On the equilibrated manifold each subsystem performs an independent symmetric walk with persistence $1 - q$, so Z is a symmetric chain with persistence $r = q^2 + (1 - q)^2$, which differs from $1 - q$ and cross-classifies the subsystem states exactly as before. Novelty is therefore satisfied on Franklin and Robertson's condition rather than on the weaker reading that a dependence is stable at the macro grain, which is the reading their binary star already excludes. Within the observed process, each macrostate is realized only by equilibrated configurations, and those are homogeneous in their onward profile, so conditional irrelevance holds strictly. An admissible perturbation that acts faster than the relaxation sets a subsystem while its hidden variable is still stale. That reaches a realization of the same macrostate in which the hidden variables disagree, and from there $P(Z_{t+1} = 0) = 1 - r$ rather than r . Closure fails.

The perturbation is admissible in the same three respects: it acts on the candidate partition through a physically available channel, it leaves the dynamics untouched, and the system re-equilibrates within a step, so the regime is preserved and the deviation is a transient.

What differs between the two cases is the source of the borrowed stability. The latent-driver proxy borrows from an upstream driver that is exogenous and persistent; hidden-feedback parasitism borrows from an internal relaxation that is fast and endogenous. The shared skeleton is the substance of the claim. One structural fact generates the gap in both, that the observed process visits only a homogeneous subset of a heterogeneous macroclass, and the two sources show how little that fact depends

on where the off-partition structure sits. A defender of the screening-off-plus-novelty package might concede the latent-driver proxy as an edge phenomenon while keeping the criterion intact. Two sources with the same signature are harder to relegate, because what is then at issue is not a peculiar configuration but a general category: apparent stability resting on structure the partition itself does not preserve under admissible perturbation.

The intervention test here is accordingly not just standard Woodwardian causal testing in new language. Standard interventionist analysis asks whether one variable causes another within an already accepted partition. The question here is prior to that: it asks whether a candidate partition itself carries autonomous transition structure under admissible perturbation.

Without an intervention-indexed test, genuinely autonomous macro-structure is indistinguishable from structure that merely inherits apparent autonomy from upstream. That is why admissibility must be declared before scoring rather than inferred from verdict patterns afterward, and it is the wedge motivating the rest of the paper.

A similar point applies to causal-emergence work. Effective-information analyses identify cases where macro-descriptions can be more causally informative than micro-descriptions, often by improving effectiveness through changes in determinism and degeneracy (Hoel et al. 2013); Dewhurst (2021) argues that such gains support only epistemic emergence absent further constraints. That diagnosis is correct as far as it goes, but it identifies a gap rather than filling it. A partition can increase effective information while remaining fragile under intervention or horizon variation. The approach here treats such gains as corroborating evidence within a closure-governed criterion, not as a standalone ontology test.

Relatedly, Wallace’s dynamic treatment of higher-level ontology is right to push the debate away from mere collections of parts and toward structure “in the distributions and dynamics” (Wallace 2024). But a commuting or phase-space friendly representation is still not yet a full admittance criterion. Closure asks when the higher-level structure holds up under admissible perturbation rather than merely being calibrated to observed trajectories. Ladyman’s recent contribution to the real-patterns volume moves in the same dynamic direction. He characterizes real patterns in terms of “projectible generalisations that predict and explain” (Ladyman 2026, p. 1). The proposal here is that what remains to be added is explicit admission and downgrade discipline.

So the paper is not computational mechanics or causal-emergence analysis with new labels. It is a philosophical criterion that borrows formal tools while adding intervention-indexed commitment conditions, an explicit commitment protocol, and a sustained argument about why closure is the right fix for pattern-realism’s permissiveness problem.

4 Non-Ideal Application: Leakiness, Diagnostics, and Verdicts

4.1 Why Approximation is Expected

In complex systems, boundaries leak and couplings shift across regimes. Exact closure is therefore unusual. A credible realism criterion cannot require perfect closure everywhere. The study of approximate aggregation has a long history, especially Simon’s account of hierarchy and near-decomposability, where strong within-subsystem couplings can coexist with weaker cross-subsystem couplings and thereby separate short-run local dynamics from longer-run aggregate behavior (Simon 1962; Ando and Simon 1961). The extension here is philosophical rather than technical: it uses graded closure to support graded ontological verdicts, not to improve estimation. Rosas et al.’s distinction between informational, causal, and computational closure is useful here because it shows why the formal story is already graded and plural at the diagnostic level even when used to motivate one ontological criterion (Rosas et al. 2024).

Approximation is the expected non-ideal form of the same criterion, provided constraints are fixed and diagnostics are comparative.

Ontology tests should be type-level before token-level. A pattern type can be robustly closed for a regime even when a specific token fails because of boundary violation, atypical perturbation, or timescale mismatch. One anomalous token is therefore not decisive. The relevant question is whether failures are exceptional or systematic for the type under the stated constraints. Systematic failure, where within-class micro-differences repeatedly matter for macro-transitions across tokens under fixed constraints, is what triggers type-level downgrade rather than token-level exception.

4.2 Leakiness and the Distribution It Is Indexed To

Leakiness measures how much within-class micro-detail still improves macro-future prediction once the current macrostate is fixed. The natural quantity is the conditional mutual information $I(X_t; Z_{t+1} \mid Z_t)$ (Cover and Thomas 2006, eqs. (2.60)–(2.61)). In plain terms, a non-zero value means that residual micro-detail still helps predict the next macrostate even after the present one is fixed, which is the signature of a leaky partition.

Stated that way the quantity is incomplete, and the reason matters for the criterion rather than only for estimation practice. A conditional mutual information is defined relative to a joint distribution, and nothing in the notation says which distribution over within-class realizations is meant. Read against the observed joint, the measure inherits exactly the limitation the paper’s central argument turns on. The latent-driver proxy of §3.2 makes the point sharply: there, screening off holds strictly, so $I(X_t; Z_{t+1} \mid Z_t)$ is exactly zero under the observed joint, while the candidate fails transition autonomy. A measure that returns perfect closure for a candidate the criterion must reject cannot be the criterion’s target quantity.

The repair is to index leakiness to the distribution over within-class realizations, which yields two quantities with different evidential roles rather than one ambiguous one.

1. Observational leakiness: $I(X_t; Z_{t+1} \mid Z_t)$ evaluated under the observed joint. This is a screening diagnostic. It is cheap, it requires no interventional access, and a non-negligible value convicts immediately, since micro-detail that demonstrably improves macro-prediction within the observed process is transition-relevant heterogeneity the partition has failed to absorb. What it cannot do is acquit. A value of zero shows only that the realizations the process happens to visit are homogeneous, which is consistent with borrowed autonomy.
2. Admissible leakiness: the same functional evaluated under the distributions in which the within-class realization is set by an admissible intervention. This is the criterion's quantity, and its domain has to be stated, since the verdict turns on it. For each macrostate z , let $R_I(z)$ be the microstates x with $g(x) = z$ that are reachable by interventions in the admissible class I while preserving the regime, and let $\mathcal{M}_I(z)$ be the finite-support distributions over $R_I(z)$ induced by randomized mixtures of admissible interventions. No single such distribution is privileged, so the quantity is a supremum over them and over macrostates, taken at the declared horizon L :

$$\Lambda_I^{(L)}(Z) = \sup_z \sup_{\mu \in \mathcal{M}_I(z)} I_\mu(X_t; Z_{t+1:t+L}),$$

where conditioning on Z_t drops out because μ is already confined to one macroclass, and the target is the L -step path rather than the next step alone, matching the criterion of §2.3. The supremum is what makes the quantity a test of the candidate: under an average, a partition could earn robust standing by being homogeneous on some convenient weighting of its realizations.

Two points about that definition need stating, since a reviewer will ask about both.

The kernels must be indexed to the interventions that produce them, not to a surgery on the whole micro-state. Writing the pairwise condition with $do(X_t = x)$ would misdescribe the admissible class and would sit badly with the restriction on Pearl-style surgery in §2.6: an admissible intervention here resets one subsystem and leaves the drivers untouched, so it never sets X_t wholesale, and not every member of $R_I(z)$ need be reachable by setting the micro-state directly. Define instead the post-intervention kernel

$$K_I(\cdot \mid x) = P(Z_{t+1:t+L} \in \cdot \mid X_t = x, \text{ admissible controls dormant thereafter}),$$

for $x \in R_I(z)$, and state closure as the requirement that $\sup_z \sup_{x, x' \in R_I(z)} D(K_I(\cdot \mid x), K_I(\cdot \mid x')) = 0$ for a divergence D separating distinct distributions.

The two formulations agree. If $\mathcal{M}_I(z)$ contains every binary mixture over $R_I(z)$, then $\Lambda_I^{(L)}(Z) = 0$ exactly when all realizations in $R_I(z)$ induce the same distribution over $Z_{t+1:t+L}$, for every z . One direction is immediate: if the kernels coincide on $R_I(z)$, then $Z_{t+1:t+L}$ is independent of X_t under any μ supported there, so the mutual information vanishes. For the other, suppose $K_I(\cdot \mid x) \neq K_I(\cdot \mid x')$ for some $x, x' \in R_I(z)$, and let μ put equal weight on the two. The conditional distribution of $Z_{t+1:t+L}$ then differs from its marginal on a set of positive measure, so $I_\mu(X_t; Z_{t+1:t+L}) > 0$ and the supremum is positive. Either form therefore vanishes just in case within-class

micro-differences do not alter the macro-transition profile, which is transition closure as stated in §2.3. In the latent-driver proxy the observational quantity is exactly zero while both admissible forms are strictly positive, which is the separation the criterion turns on.

The two coincide whenever the observed process already visits the reachable realizations of each macrostate with appreciable weight, which is why observational diagnostics are so often adequate in practice and why the literature has had no occasion to separate them. They come apart exactly in the parasitism cases, where the observed distribution’s support is confined to a homogeneous subset of a heterogeneous class.

This also puts the exact benchmark in its proper place rather than leaving it as a parallel formalism. Strong lumpability, in Kemeny and Snell’s sense, requires that the coarse-grained process be Markov for *every* initial distribution, which is equivalent to requiring equal onward macro-transitions for every microstate in a class (Kemeny and Snell 1960). That is the limiting case in which admissible leakiness vanishes for every distribution supported on a macroclass. What observational leakiness can certify is weaker, and the contrast is best put without leaning on a term whose formulations vary with the initial distribution assumed: the observed process induces a Markov macro-process for its actual initial or stationary distribution, while the partition fails strong lumpability over the full microstate space. The proxy is exactly such a case, and the two leakiness quantities separate on it for that reason.

The asymmetry between the two is the same one recorded in §2.9: predictive adequacy can persist where interventional stability fails, so predictive evidence is important but not final. Nothing here demotes observational leakiness to irrelevance. It remains the first thing to compute and the cheapest way to eliminate candidates. The claim is only that it screens rather than settles, and that a robust verdict requires the admissible quantity.

Using information-theoretic measures to state an ontological criterion might appear to assume a computationalist metaphysics. It does not. Both quantities are functionals of transition kernels, and the conditional independencies they measure hold or fail as facts about those kernels, whatever anyone computes and whether or not any system in the world exploits them (§2.10). No claim about the cost of tracking micro-detail, thermodynamic or otherwise, is needed to make the measure objective.

Admissible leakiness is the target quantity, and observational leakiness is its most accessible screen. Other diagnostics, such as within-class transition divergence and predictive gain from added micro-features, function as estimators or proxies when direct estimation is limited, and they inherit the same indexing: what they are evidence for depends on whether the within-class realizations they range over are those the process visits or those an admissible intervention can reach.

Leakiness should therefore be read comparatively and through robustness checks, not as a context-free absolute cutoff. Absolute tolerance is domain-relative, but high sensitivity of leakiness rankings to modest defensible modeling choices is itself a downgrade trigger.

Comparative ranking is necessary but not sufficient, since a candidate can beat nearby alternatives and still fail the criterion if absolute leakiness remains high and

instability persists under modest robustness checks. Both relative superiority and minimum viability are required.

The viability floor can be stated without a numeric threshold. A partition fails minimum viability when, across the declared robustness checks, within-class micro-features yield consistent, non-negligible gains in macro-future prediction or intervention response relative to the best macro-only model. Persistence of that pattern across diagnostics and perturbation tests is decisive: the partition has not achieved the transition autonomy the criterion requires, regardless of how it ranks against competitors.

The term “non-negligible” can be given a comparative anchor without fixing a universal numeric threshold. A within-class gain is non-negligible when its magnitude is large enough that, were the same gain observed between two candidate macro-partitions, it would be recorded as a genuine comparative difference under the same diagnostic conventions. The viability floor therefore uses the within-class gain as a within-system comparandum rather than an external threshold: a partition fails if adding micro-features produces the kind of effect size that would itself count as diagnostic evidence against a rival candidate. Baseline model class and gain metric are fixed upstream along with regime and admissibility under the canonical selection discipline (§4.3). The viability floor is applied first as a minimum gate; comparative ranking then assigns verdict categories among candidates that pass it.

Approximate closure is therefore not a discretionary replacement for exact closure. It is the non-ideal evidential route to the same ontological target. Judgment is constrained by upstream fixing of regime, admissibility, baseline model class, and gain metric, by comparative stability across modest robustness checks, and by explicit downgrade or indeterminacy when those rankings do not stabilize.

Verdict-category boundaries carry genuine vagueness because the underlying property of transition autonomy is itself graded. This is a feature rather than a defect: biological fitness is real and explanatorily significant without admitting a sharp boundary between fit and unfit. What matters primarily is ordering stability. When partition *A* consistently shows lower leakiness and greater perturbation stability than partition *B* across diagnostics, that ordering should be robust even if the exact placement of the boundary between “qualified” and “robust” carries some domain-conventional element. A partition that dominates alternatives across diagnostics and remains stable under modest changes to horizon, intervention distribution, and robustness checks earns the stronger verdict regardless of where one draws the categorical line.

4.3 Procedural Safeguards in Non-Ideal Cases

Approximate closure claims require explicit safeguards. The canonical selection discipline for the criterion, referenced throughout the paper, is as follows.

1. Fix regime, horizon, admissibility class, diagnostics, baseline model class, and gain metric before comparative scoring.
2. Compare candidate partitions under the same fixed setup.
3. Test robustness under modest changes in horizon and intervention distribution.

4. Apply a minimum-viability floor: if all candidates remain high-leak and fragile, return no robust standing verdict for that regime.
5. Report the closure profile and the evidential grade separately, then the verdict they warrant (§4.8).
6. Disclose any post hoc revision to the setup and restart the evaluation rather than silently inheriting prior results.

These safeguards also fit Wimsatt’s robustness-analysis idea: confidence should rise when partially independent ways of detecting the same structure converge, and it should fall when apparent agreement is driven by shared assumptions or disappears under small perturbation (Wimsatt 1981). Multiple diagnostics do not replace criterion-level argument with a vote count. They reduce the risk that a favorable verdict is an artifact, or pseudorobust in Wimsatt’s sense, of one estimator, one horizon choice, or one admissibility specification.

This keeps threshold choice procedural rather than discretionary. It also addresses a predictable confusion. Estimation difficulty is an epistemic limit on access, not a defect in the metaphysical criterion itself. What is graded in non-ideal cases is not confidence alone, but the stability of closure itself across regime, horizon, admissibility, and robustness variation.

The purpose of this discipline is to make judgment auditable, not to eliminate it, and to prevent favored partitions from being protected by moving criteria.

4.4 Regime Dependence and Projectibility

Regime dependence is a claim about actual transition structure under specified constraints, not about observer-relativity. A pattern can be closed in one regime and leaky in another because the structure has changed, or because the available control channels have shifted. Since the admissible intervention class is defined by the regime’s transition structure (§2.6), a change in that structure can alter both the admissible class and the closure verdict even when the system’s micro-dynamics remain the same.

The metaphysical stance is explicit: regime and horizon index the modal profile being claimed, not a concession that anything goes.

An ordinary physical analogy helps. Ice crystals are objective structures only within certain temperature and pressure regimes. That regime dependence shows that objective structure can be conditional on equally objective background constraints without becoming epistemically subjective.

Projectibility provides a further check. Defensible regime specifications should support stable induction under modest counterfactual variation. Narrowly engineered regimes that protect a favored partition usually fail under small context shifts. When that occurs, the candidate was never robustly closed in the relevant sense.

Here the paper is closest to Ladyman and Ross. Projectibility is not treated as an optional methodological virtue. It functions as a realism-relevant stress test on whether a closure claim survives beyond calibration conditions (Ladyman and Ross 2007). Closure without projectible robustness is too cheap to support robust realist claims.

This is also why regime-indexing is not a concession to weakness. It is what allows robust verdicts in the special sciences to be disciplined rather than impossible. Without explicit regime and intervention indexing, most biological and social kinds would look merely leaky in an undifferentiated way. With that indexing in place, the question becomes whether the candidate is transition-autonomous within the bounded domain that gives it its explanatory role.

But projectibility is still not enough by itself, because a pattern can remain projectible while borrowing its stability from upstream structure (see §3.2). Closure asks the further question of whether the candidate partition carries its own transition structure under admissible perturbation.

Ladyman and Lorenzetti’s effective ontic structural realism sharpens the same point at the metaphysical level. If ontology is regime-dependent and scale-relative, then the question is not whether a candidate belongs to a single final inventory of being, but whether it is stable enough in the relevant regime to merit commitment there (Ladyman and Lorenzetti 2024). Closure is offered here as the operational discipline for making that judgment explicit.

Regime dependence itself should be split into two forms. Ontic regime dependence occurs when system structure changes, such as phase transitions or boundary-condition shifts. Epistemic regime dependence occurs when measurement or control access changes while underlying structure remains fixed. The first changes what is there to be tracked. The second changes what can be warranted from available evidence.

4.5 Failure Conditions and Downgrade Rules

The account should also say clearly when commitment should be withdrawn or downgraded. Four failure patterns do that work.

1. Persistent high leakiness across defensible diagnostics under fixed constraints.
2. Strong disagreement among diagnostics that does not resolve under modest robustness checks.
3. High sensitivity of verdicts to small, defensible changes in admissibility or horizon.
4. Candidate sets where no partition clears the minimum-viability floor.

When these patterns appear, the right response is not forced binary judgment. The right response is explicit downgrade to qualified or indeterminate status. This keeps the criterion resilient without pretending that every case must produce a sharp ontology verdict.

Conversely, when closure persists across distinct, independently motivated admissibility classes within the same regime, that cross-admissibility stability raises evidential confidence for robust commitment.

Type-token reminder: a downgraded token does not automatically refute type-level closure. The key question is whether instability is exceptional at token level or systematic for the type under the stated constraints.

4.6 Stable Versus Merely Entailed Patterns

The graded account also supports an important distinction. A pattern can be entailed by microhistory and laws without being stable as a macro-handle. Entailment alone is cheap; stability under admissible perturbation is demanding. The distinction matters for permissiveness debates, because a contrived disjunctive construction can be true of a realized trajectory while still failing closure: it can require continual within-class micro repair, fail under modest regime shifts, and lose interventional reliability.

A historical case makes the pattern concrete. Phlogiston theory compressed combustion phenomena under a single macro-variable, but the partition required persistent bookkeeping to survive: when metals gained weight upon burning, theorists introduced “negative phlogiston”; when different substances showed different weight changes, further ad hoc parameters appeared. Each patch was a new within-class distinction imported to preserve macro-prediction. Phlogiston was, in the vocabulary of §3.1, a high-maintenance partition: reliable macro-prediction repeatedly forced refinement of within-class distinctions. By contrast, oxygen theory required substantially less such repair: the macro-variable (oxidation state) screened off the relevant chemistry without persistent bookkeeping. The critique here is structural, not retrospective ridicule. A candidate that continually imports corrections to preserve macro-prediction behaves like a non-autonomous partition, and the closure framework detects this failure directly. The distinction between principled theoretical development and ad hoc rescue is itself contested philosophy of science, and the illustration is reconstructive rather than a clean empirical test. Closure supplies one principled way to draw the distinction, namely whether the supplementation requires persistent within-class micro-bookkeeping or introduces new transition-autonomous structure.

So gerrymandered constructions are not necessarily false; they usually fail to qualify for robust macro-level commitment. They may remain descriptions, but not ontologically weighty descriptions in the relevant regime. The same applies to computationally adequate models more broadly. A higher-level model can be useful for prediction or control without meeting closure standards. Computational adequacy means the model serves some purpose; closure requires that transition-relevant micro-differences are screened off in the specified regime. A computationally adequate but high-leak representation remains a valuable tool, but it does not automatically earn realist commitment. Closure is not a synonym for “good model.” It tests whether the relevant transition structure is in the world at the target grain, rather than merely encoded in a successful representation.

4.7 Practical Qualification Under Sparse Intervention Access

In many domains, direct interventions are sparse, ethically constrained, or expensive. Sparse intervention access changes evidential strategy, not the criterion itself. The evidential order is asymmetric: predictive success can raise confidence, but only interventional stability can underwrite robust commitment.

Where intervention data are limited, predictive closure functions as an operational indicator while interventional closure remains the ontological target. A model can fit historical trajectories and still fail under novel perturbation. For that reason, claims

should be qualified by available intervention access and downgraded when out-of-regime behavior is unstable.

In-sample prediction is therefore not treated as decisive. The standard remains robustness under admissible perturbation, even when that robustness must be assessed indirectly.

Relevant evidence need not come only from humanly executable interventions. In observational sciences it can also come from naturally occurring disturbances, comparative cases, exogenous shocks, policy changes, and other perturbations that bear on the same transition structure. Such indirect intervention evidence often reveals whether a macro-description remains stable when the system is pushed off its ordinary trajectory.

This point also explains why diagnostics should be triangulated. Predictive closure can look strong in-sample while interventional closure fails under admissible perturbation. Partial observability can hide within-class heterogeneity that later appears as instability. Nonstationarity can make a previously low-leak partition unstable outside its calibration window. Triangulation does not remove these risks, but it makes them visible.

Payment systems are a useful case, since direct intervention on them is rare and mostly not ours to perform. A well-functioning payment system can look observationally stable for long stretches. What matters for stronger commitment is how that stability behaves under settlement delays, enforcement shocks, trust disruptions, or policy changes that function as admissible perturbations. Where those perturbations leave transactional organization intact, robust verdicts become plausible. Where the same structure survives only observationally, or collapses once such pressures appear, the right verdict is qualified rather than robust.

4.8 Verdict Rule and Reporting Discipline

Verdicts have been running together two things that come apart, and separating them makes the rule both cleaner and more honest. Whether a candidate is transition-autonomous is a question about the candidate. Whether we have established that is a question about our evidence. A verdict is what the two together warrant.

1. *Closure profile*, a property of the candidate under the declared regime, horizon, and admissible intervention class. *Closed* when admissible leakiness is at or near zero; *partially closed* when it is low within the declared constraints but rises under modest variation of them; *high-leak* when within-class micro-differences materially alter the macro-transition profile. Nothing here is about what anyone knows.
2. *Evidential grade*, a property of our access to that profile. *Strong* when interventional evidence spans the admissible class and diagnostics converge; *limited* when the evidence is largely predictive because intervention access is sparse, costly, or ethically closed (§4.7); *unstable* when diagnostics disagree persistently or rank-orders do not settle.
3. *Commitment verdict*, the attitude the first two warrant. *Robust* for a closed profile established on strong evidence. *Qualified* for a partially closed profile on strong evidence, or a closed profile on limited evidence, since fragility in the candidate

and thinness in the evidence both justify commitment short of robust, for different reasons. *Indeterminate* when the evidential grade is unstable, whatever the profile may be. The name marks our position, not the candidate's: what is indeterminate is the verdict, not the transition structure it is about.

The separation earns its keep immediately, because it exposes a case the old vocabulary could not express. A high-leak profile on strong evidence is not an indeterminate verdict. It is a determinate negative: the candidate has been shown to lack transition autonomy, and robust standing is denied rather than deferred. Collapsing that into indeterminacy would report a finding as a failure to find. Conversely, an indeterminate verdict says nothing about the candidate. A candidate can be perfectly closed and still receive one, and better evidence may later show that it was.

The rule is minimal rather than maximal: stricter domain conventions can narrow the robust category further, but nothing weaker than this triangulation should underwrite robust verdicts.

Where the canonical selection discipline (§4.3) fixes *what is held constant* before scoring, reporting fixes *what is disclosed* once scoring is done. To keep conclusions comparable across cases, closure assessments should state five items explicitly: target partition and rationale; regime and horizon specification; admissible intervention class and justification; diagnostics used and their agreement or disagreement; and final verdict category. This disclosure structure prevents silent shifts in target, timescale, or admissibility from being mistaken for genuine ontological progress.

Reports should therefore carry the profile and the grade, not just the verdict, since the verdict alone does not say which of them drove it. A qualified verdict on a fragile candidate and a qualified verdict on a well-behaved candidate we have barely probed are different situations with different remedies: the first calls for narrowing the regime, the second for better evidence. That is also why the anti-instrumentalist claim of §2.10 attaches to the profile rather than to the verdict. The profile is a fact about the candidate. The verdict is an attitude, and an attitude is answerable to evidence as well as to the world.

Diagnostics do not replace criterion-level argument. They provide comparative evidence about whether a candidate satisfies the criterion under declared constraints. No single proxy is treated as infallible. Three checks are usually sufficient in practice: a leakiness proxy based on predictive gain from added within-class micro-features; within-class transition-profile divergence; and intervention-response invariance under admissible perturbation. Agreement across these checks increases warrant; persistent disagreement is evidence for revision or downgrade rather than for forced commitment. When interventional evidence is sparse or unavailable, the remaining diagnostics can still support qualified status, but robust standing requires that the interventional leg not be entirely missing.

5 Objections and Replies

Two objections are taken here. A third, that closure merely renames screening off properly understood, was answered in §2.2, where the concession is made directly. Three standing commitments apply to both replies and are not restated in each. First,

the account does not deny microphysical completeness; it claims non-redundancy at the explanatory and interventional level for declared macro-targets, so the causal-exclusion pressure is met by target-indexing rather than by multiplying causes. Second, all verdicts are indexed to the declared constraints, so objections presupposing context-free ontological claims address a view the paper does not hold. Third, the strongest conclusion is conditional on structural realist and interventionist commitments; readers who withhold those commitments can still accept the narrower anti-gerrymandering result.

5.1 “Instrumentalism and Admissibility”

Objection: target selection is interest-relative, so the view still collapses into usefulness.

Reply: target selection can be interest-shaped without making closure verdicts preference-shaped. Once regime, horizon, and admissibility are fixed, whether macro-transitions are autonomous is a system fact under explicit constraints.

The stronger version invokes Dennett’s Martian (Dennett 1991, p. 42): if an ideal micro-predictor can forecast everything, macro realism looks optional. But this infers ontological redundancy from representational power. Even a micro-omniscient predictor coexists with objectively better macro-bottlenecks for target dynamics, and ignoring those bottlenecks increases representational burden without erasing macro transition structure.

Observer power changes convenience, not closure facts. Whether a partition is transition-autonomous under fixed constraints does not vary with who computes it.

Admissibility version of the same objection: admissibility criteria already encode what counts as the relevant level, so the circularity worry returns at one level up. The consolidated reply is that admissibility selection is not a free parameter. It is constrained by the system’s actual control ecology, which is to say the physically realizable, regime-preserving perturbations the system already faces under thermal fluctuation, diffusion, mechanical contact, and similar naturally occurring sources. The analyst’s task is to identify that ecology, not to construct it. Hierarchical order and disclosure rules make the identification auditable: admissibility is fixed from physical constraints before partition scoring, and post hoc admissibility revision triggers restart. Where the ecology is genuinely contested, the criterion correctly returns indeterminate verdicts rather than forcing a favored outcome. The worry that admissibility “shifts” the hard problem treats intervention-class choice as free; the framework treats it as a physically anchored feature of the regime being evaluated. Shifting is what happens when one problem is replaced by an equally hard one. Here the replacement is by a problem that is at least partly settled by the regime itself, since which control channels exist is not something the analyst decides. That does not make the choice automatic, and it is not claimed to.

Residual disagreement can remain. The account does not promise universal convergence from one pass. It promises disciplined comparison and explicit downgrade when verdicts are unstable across nearby defensible admissibility specifications. This is a feature for skeptical readers: hard cases are flagged as unresolved rather than settled by stipulation.

Instrumentalist pressure hides in two forms: ontology should track predictive utility only, or ontology is unnecessary once predictive utility is available. Both are wrong. Utility without closure is too permissive, and closure without ontology understates what stable intervention-guiding structure provides. What closure adds is commitment discipline, not metaphysical extravagance. If a candidate repeatedly supports stable counterfactual control under fixed constraints, withholding ontological standing is a position that owes an account of what further condition realism was supposed to require. Rather than forcing maximal realism, the paper asks for explicit criteria for when anti-realism remains credible.

The strongest anti-instrumentalist point is structural rather than rhetorical. The objection that leakiness is “only” predictive error has real force against the observational quantity, which is why that quantity screens rather than settles (§4.2). Directed at admissible leakiness the objection presupposes a gap between tracking causal structure and describing what is real. But under the paper’s structural realist commitments, there is no further ontological layer for that gap to open onto. Stable counterfactual and predictive organization is not evidence for some deeper condition left unspecified elsewhere; it is what the relevant reality consists in at that grain. A partition that leaks under admissible intervention is not one where we failed to see the real object hiding underneath. It is one where the candidate lacks autonomous transition structure. Admissible leakiness therefore measures ontic deficiency in the candidate partition, not merely our ignorance. Once the test and the target are properly separated, the prediction miss is itself the structural fact.

The cheap-coding objection fails for the same reason. A recoding can improve local fit and still fail closure under admissible perturbation. The criterion is therefore representation-hostile in exactly the way permissiveness critiques demand.

5.2 “Closure Is Too Strong and Eliminates Most Special Sciences”

Objection: if closure requires that within-class micro-differences not matter for macro-transitions, then most special-science kinds will fail. Biological kinds are notoriously leaky (Boyd 1991), and many psychological states and economic categories are no sharper. The criterion eliminates the ontology it was supposed to discipline.

Reply: the objection conflates robust standing with any positive verdict at all. Three verdict categories are available, not one. Robust standing requires stable closure under fixed constraints. Qualified standing fits cases where closure is partial, regime-limited, or supported by convergent but incomplete evidence. Indeterminate status fits cases where evidence is too unstable to warrant commitment.

Most special-science kinds fall into the qualified range, and that is the correct result. Biological species exhibit substantial transition autonomy within ecological and evolutionary regimes while leaking at boundary cases (hybrid zones, ring species, horizontal gene transfer). Psychological kinds often support predictive and interventional regularity within clinical or experimental regimes while failing under broader perturbation. The criterion does not eliminate these kinds. It gives them the status their closure evidence supports: qualified standing with explicit regime limitations, rather than either full robust status or wholesale elimination.

A common misunderstanding is that exact transition autonomy must hold everywhere if a special-science kind is to count as robustly real at all. That is too strong. Because the admissible intervention class helps define the regime, a macro-variable can achieve robust standing within a narrow but physically bounded ecological, physiological, or institutional home even if it leaks outside that home. A biological trait or economic variable need not be universally transition-autonomous to be robustly real. It need only be transition-autonomous under the intervention class that defines the regime in which it operates. Downgrade to qualified status becomes appropriate when those regime boundaries are stretched, or when intervention evidence within the regime remains too sparse to warrant the stronger verdict.

Nearby real-pattern work in biology provides a useful foil. One can characterize biological objects by their explanatory power and pragmatic utility, and by the availability of multiple valid carvings, without yet having shown that lower-level variation is transition-irrelevant in the stronger sense used here (Jiang 2025). Such utility matters, but it does not by itself settle robust standing.

A sharper form of the objection concedes the graded verdicts and targets their distribution: the criterion is decisive where it is easy, in declared-channel settings like the Markov benchmark, and returns qualified or indeterminate verdicts almost everywhere the debate matters, so its bite is inversely proportional to its applicability. Three points blunt this. First, qualified verdicts are comparative results, not silence: they rank candidate partitions, locate where leakiness lives, and say what evidence would upgrade the verdict, which is more than a permissive criterion says anywhere. Second, rival criteria do not perform better where closure verdicts are hard; they issue robust verdicts there, and the wedge cases show that some of those verdicts certify borrowed autonomy. A criterion that withholds robust standing where the evidence cannot support it is not weaker than one that grants it there. Third, decisiveness tracks the maturity of a domain's control ecology rather than a boundary fixed in advance: as intervention practice accumulates, the same criterion delivers stronger verdicts, so its reach grows with experimental practice instead of being confined to clean cases by construction.

The objection therefore proves too much. A criterion that granted robust standing to every special-science kind regardless of leakiness would be the permissive view closure is designed to replace. The gain from graded verdicts is that they match evidential reality rather than forcing a binary choice between full realism and eliminativism.

6 Conclusion: A Disciplined Realism Claim

The three claims distinguished at the outset rest on different things. The anti-gerrymandering constraint rests on the wedge cases of §3.2 and requires no particular realism package. The criterion for robust commitment rests on the structural-realist and interventionist commitments of §1.2. The structured-space reading rests on neither, and is offered as a direction rather than a result: a reader who grants the first two can withhold it without loss to the admittance-and-downgrade protocol.

Rainforest realism is compelling, but its admittance criteria remain incomplete. Screening off plus novelty remains necessary, but observational success can be parasitic: the latent-driver proxy and hidden-feedback cases (§3.2, §3.2.2) show that a candidate can satisfy observational screening off while borrowing its apparent autonomy from off-partition organization the partition itself does not preserve under admissible perturbation. Closure under admissible intervention is what distinguishes autonomous from borrowed macro-structure. Strong lumpability provides the exact benchmark in Markov settings. Leakiness and convergent diagnostics extend the same criterion to non-ideal settings as a graded verdict protocol with explicit downgrade discipline.

This further step sharpens the non-redundancy demand already implicit in the Primacy of Physics Constraint (§2.5): a candidate is non-redundant when lower-level variation within its macroclasses stops mattering to the relevant macro-transition profile under declared constraints. In the recent literature, the criterion supplies what Franklin and Robertson’s screening-off test, Rosas et al.’s formal diagnostics, Ladyman and Lorenzetti’s effective realism, and recent dynamic formulations of pattern realism do not yet by themselves supply: an explicit closure-based admittance and downgrade discipline for robust macro-level commitment (Franklin and Robertson 2024; Rosas et al. 2024; Ladyman and Lorenzetti 2024; Wallace 2024; Ladyman 2026).

The proposal supports selective commitment: robust where closure is stable, qualified in borderline cases, explicitly downgraded where verdicts depend on fragile admissibility or horizon choices. The overall position is intentionally middle-range, stronger than compression-only pattern realism because it adds exclusion and downgrade structure, weaker than a total-level metaphysics because it stays restricted to induced closure in spatiotemporal systems under declared constraints.

Separating what is established from what is offered programmatically keeps the contribution accurately calibrated. The anti-gerrymandering role of closure, and the selector role that operationalizes the modal reading of screening off, stand on the argument of Sections 2 through 4. The further reading of rainforest realism as a structured space of computationally closed coarse-grainings is offered programmatically, in line with the three claims distinguished in §1. Under that programmatic reading, the verdict categories make the structure reportable: robust where placement is stable, qualified where ontological support is real but fragile, and indeterminate where no stable placement has yet been secured. The formal richness of that space in any given domain depends on what the closure discipline turns up there, and the core argument does not depend on any particular such finding.

The view separates itself from eliminativist pressure by neither treating higher-level descriptions as fictions by default nor granting them standing by convenience alone. The test is whether they track stable transition structure under admissible intervention. If they do, realist commitment follows on the paper’s stated terms. If they do not, realism should be downgraded rather than protected by rhetoric.

This paper does not offer a complete metaphysics of levels: the scope stays restricted to induced closure in spatiotemporal systems under declared constraints. It does not settle general debates about emergence or reduction; closure is one necessary

filter within a cumulative admittance protocol, not a standalone theory of higher-level reality. It supplies the dynamical component of an account of objecthood and leaves room for complementary accounts of how closed structure is realized, classified, and individuated (§1.3). It does not supply a universal implementation recipe: applications beyond the illustrations depend on domain-specific estimator choices, robustness checks, and control-channel inventories that remain downstream of the philosophical argument and do not alter its content. And the strongest conclusion is conditional on the structural realist and interventionist commitments stated in §1.2 and §2.10; readers who withhold those commitments can still accept the narrower anti-gerrymandering result. Its contribution is a philosophical criterion with application discipline, not a panacea.

Future work should test the staged admittance protocol across domains by comparing candidate partitions under fixed constraints, rather than by multiplying new concepts. The paper has made its commitments explicit and addressed worries about circularity, instrumentalist drift, and scope ambiguity directly. If those replies fail, they fail on the merits of the stated argument rather than because those issues were left silent. One further question, left for future work, is whether this closure-based account of realist commitment bears on substrate-independence or on more abstract forms of structural standing.

References

- Ando A, Simon HA (1961) Aggregation of variables in dynamic systems. *Econometrica* 29(2):111–138. <https://doi.org/10.2307/1909285>
- Bishop RC, Atmanspacher H (2006) Contextual emergence in the description of properties. *Foundations of Physics* 36(12):1753–1777. <https://doi.org/10.1007/s10701-006-9082-8>
- Boyd R (1991) Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies* 61(1-2):127–148. <https://doi.org/10.1007/BF00385837>
- Cover TM, Thomas JA (2006) *Elements of Information Theory*, 2nd edn. Wiley-Interscience, Hoboken, NJ. <https://doi.org/10.1002/047174882X>
- Craver CF (2007) *Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience*. Oxford University Press, Oxford. <https://doi.org/10.1093/acprof:oso/9780199299317.001.0001>
- Dennett DC (1991) Real patterns. *The Journal of Philosophy* 88(1):27–51. <https://doi.org/10.2307/2027085>
- Dewhurst J (2021) Causal emergence from effective information: Neither causal nor emergent? *Thought: A Journal of Philosophy* 10(3):158–168. <https://doi.org/10.1002/tht3.489>

- Dupré J, Nicholson DJ (2018) A manifesto for a processual philosophy of biology. In: Nicholson DJ, Dupré J (eds) *Everything Flows: Towards a Processual Philosophy of Biology*. Oxford University Press, Oxford, p 3–45
- Franklin A, Robertson K (2024) Emerging into the rainforest: Emergence and special science ontology. *European Journal for Philosophy of Science* 14(4):61. <https://doi.org/10.1007/s13194-024-00622-4>
- French S (2014) *The Structure of the World: Metaphysics and Representation*. Oxford University Press, Oxford
- Goodman N (1983) *Fact, Fiction, and Forecast*, 4th edn. Harvard University Press, Cambridge, MA. Originally published 1955
- Haugeland J (1998) *Having Thought: Essays in the Metaphysics of Mind*. Harvard University Press, Cambridge, MA
- Hoel EP, Albantakis L, Tononi G (2013) Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences* 110(49):19790–19795. <https://doi.org/10.1073/pnas.1314922110>
- van Inwagen P (1990) *Material Beings*. Cornell University Press, Ithaca, NY
- Jaworski W (2016) *Structure and the Metaphysics of Mind: How Hylomorphism Solves the Mind-Body Problem*. Oxford University Press, Oxford
- Jiang Y (2024) The metaphysics of mechanisms: An ontic structural realist perspective. *Synthese* 204(1):32. <https://doi.org/10.1007/s11229-024-04684-y>
- Jiang Y (2025) Biological object as real patterns: Reconciling processualism and scientific realism. *Synthese* 206(4):191. <https://doi.org/10.1007/s11229-025-05285-z>
- Kemeny JG, Snell JL (1960) *Finite Markov Chains*. Van Nostrand, Princeton, NJ. Reprinted 1976, Springer
- Ladyman J (2026) Patterns all the way up: Prolegomena to a future naturalised metaphysics. In: Millhouse T, Petersen S, Ross D (eds) *Dennett’s Real Patterns in Science and Nature*. MIT Press, Cambridge, MA. Preprint circulated in 2024; final version in edited volume
- Ladyman J, Lorenzetti L (2024) Effective ontic structural realism. *The British Journal for the Philosophy of Science* <https://doi.org/10.1086/729061>
- Ladyman J, Ross D (2007) *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press, Oxford
- Lewis D (1986) *On the Plurality of Worlds*. Blackwell, Oxford

- Machamer P, Darden L, Craver CF (2000) Thinking about mechanisms. *Philosophy of Science* 67(1):1–25. <https://doi.org/10.1086/392759>
- Meyer A (2026) Dependence, novelty, autonomy and the emergence of spacetime. *Synthese* 207(6):261. <https://doi.org/10.1007/s11229-026-05634-6>
- Millhouse T (2022) Really real patterns. *Australasian Journal of Philosophy* 100(4):664–678. <https://doi.org/10.1080/00048402.2021.1941153>
- Pearl J (2009) *Causality: Models, Reasoning, and Inference*, 2nd edn. Cambridge University Press, Cambridge
- Rosas FE, Geiger BC, Luppi AI, Seth AK, Polani D, Gastpar M, Mediano PAM (2024) Software in the natural world: A computational approach to hierarchical emergence. arXiv preprint arXiv:2402.09090. <https://doi.org/10.48550/arXiv.2402.09090>, URL <https://arxiv.org/abs/2402.09090>
- Shalizi CR, Crutchfield JP (2001) Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics* 104(3-4):817–879. <https://doi.org/10.1023/A:1010388907793>
- Simon HA (1962) The architecture of complexity. *Proceedings of the American Philosophical Society* 106(6):467–482. <https://doi.org/10.2307/985254>
- Thomasson AL (2007) Artifacts and human concepts. In: Margolis E, Laurence S (eds) *Creations of the Mind: Theories of Artifacts and Their Representation*. Oxford University Press, Oxford, p 52–73
- Wallace D (2024) Real patterns in physics and beyond. URL <https://philsci-archive.pitt.edu/id/eprint/23888/>, preprint
- Wimsatt WC (1981) Robustness, reliability, and overdetermination. In: Brewer MB, Collins BE (eds) *Scientific Inquiry and the Social Sciences*. Jossey-Bass, San Francisco, p 124–163
- Wimsatt WC (2000) Emergence as non-aggregativity and the biases of reductionisms. *Foundations of Science* 5(3):269–297. <https://doi.org/10.1023/A:1011342202830>
- Woodward J (2015) Methodology, ontology, and interventionism. *Synthese* 192(11):3577–3599. <https://doi.org/10.1007/s11229-014-0479-1>
- Woodward J (2021) *Causation with a Human Face: Normative Theory and Descriptive Psychology*. Oxford University Press, Oxford. <https://doi.org/10.1093/oso/9780197585412.001.0001>